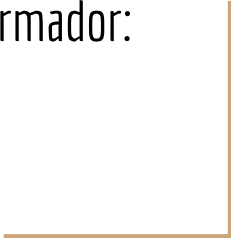


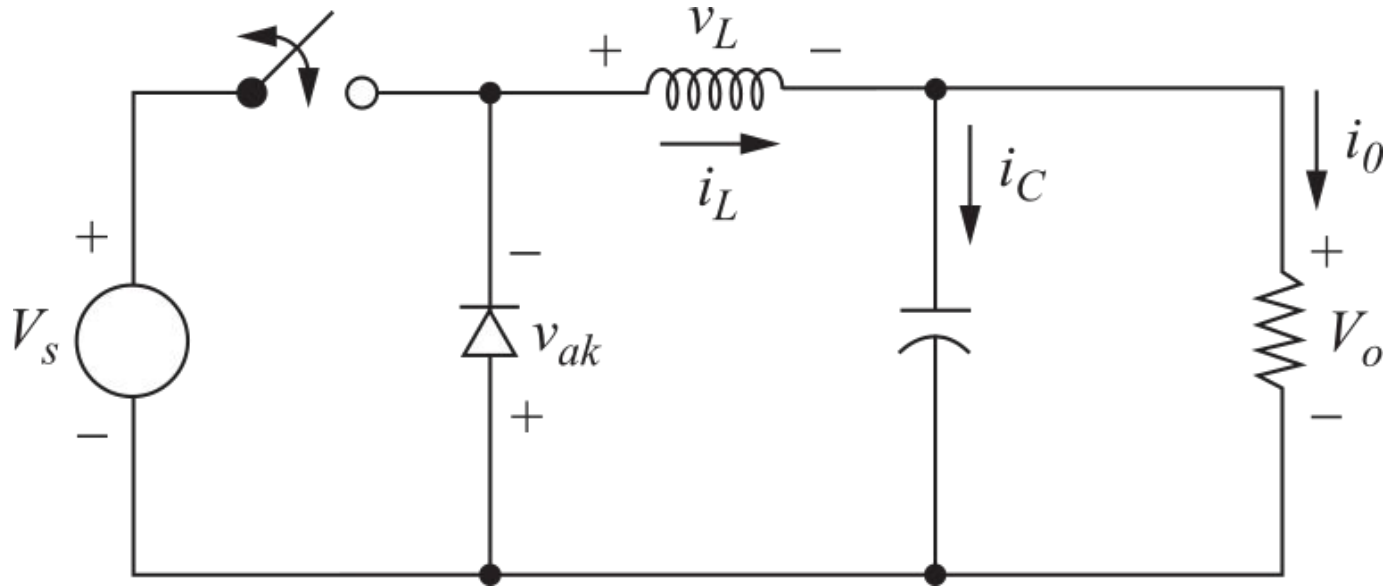


Conversión CC/CC

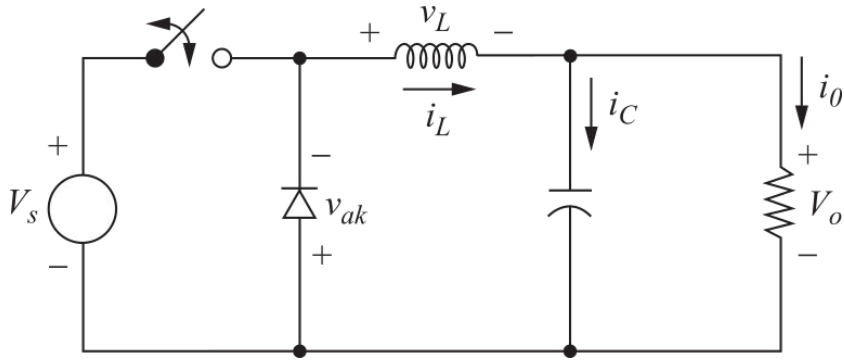
Convertidores sin Transformador:
Reductor



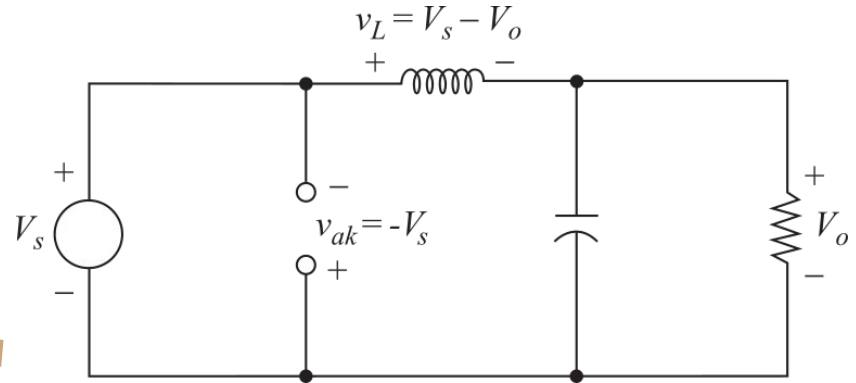
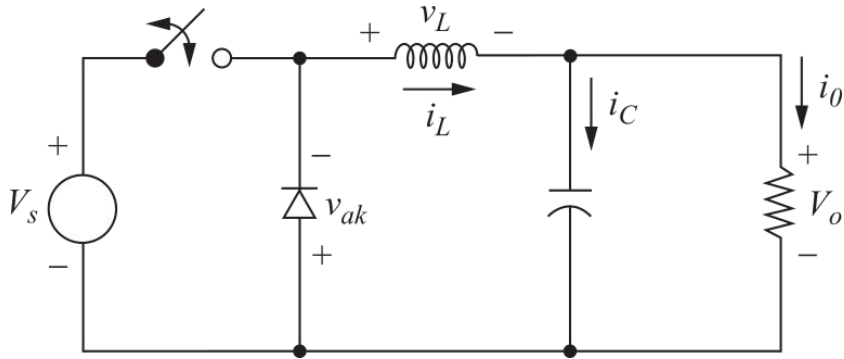
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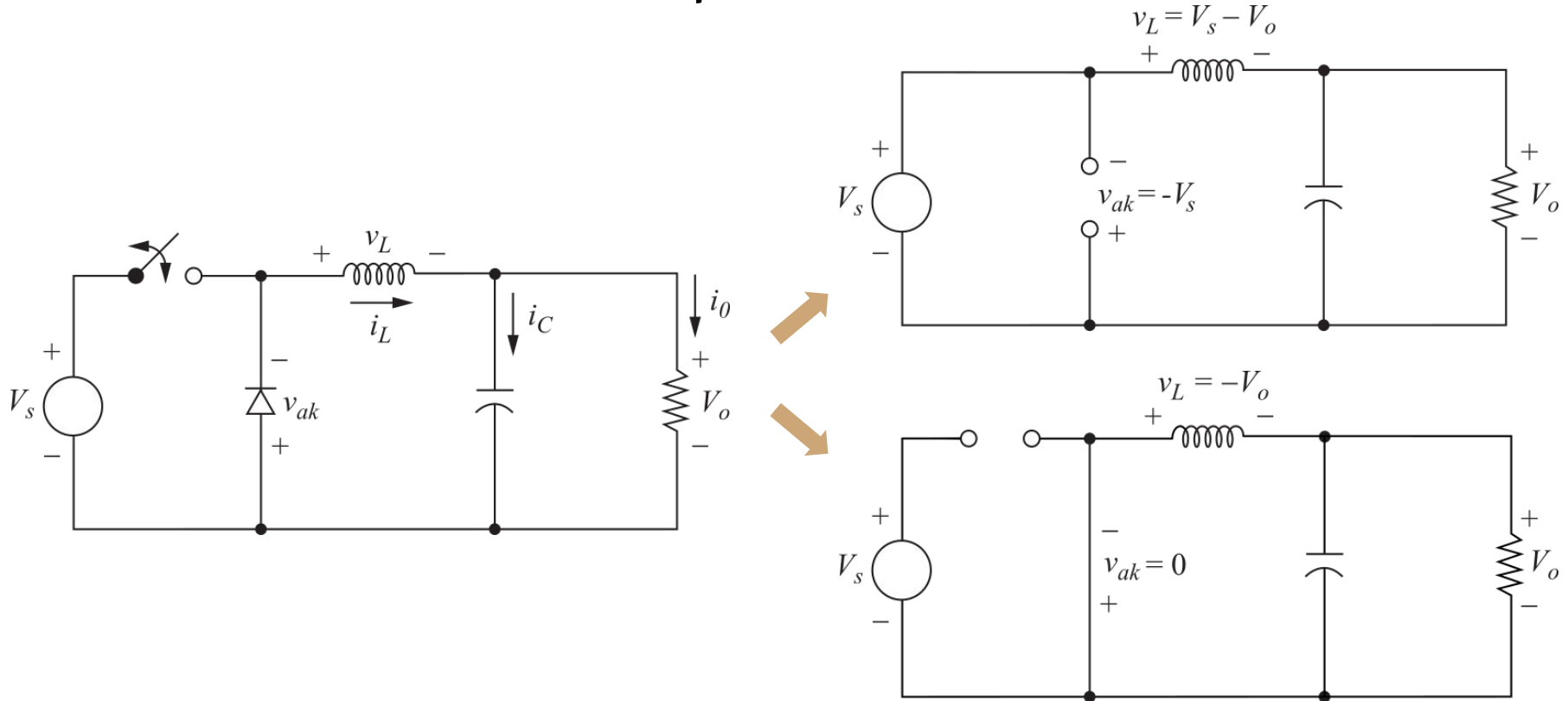
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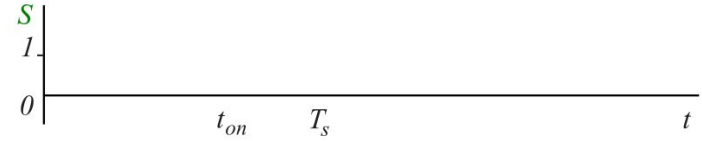
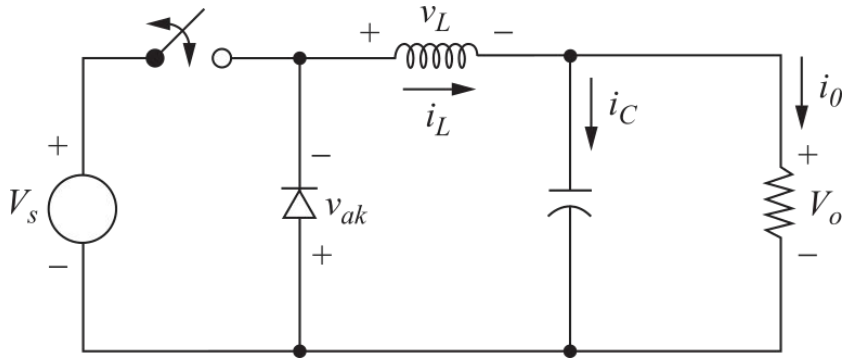
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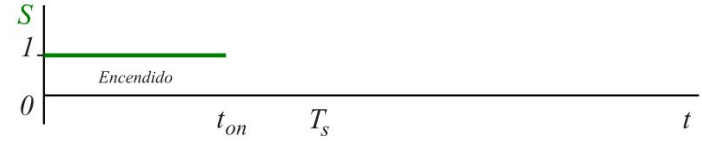
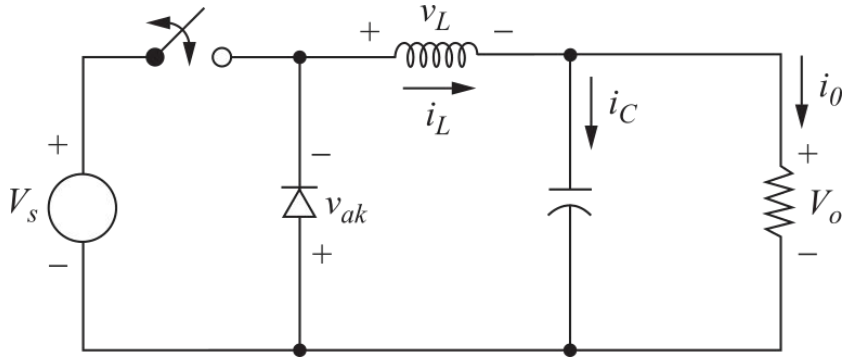
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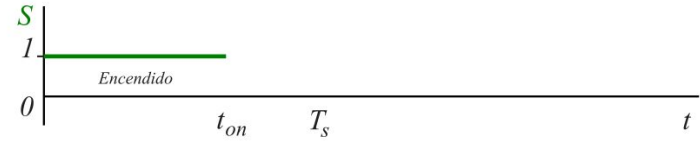
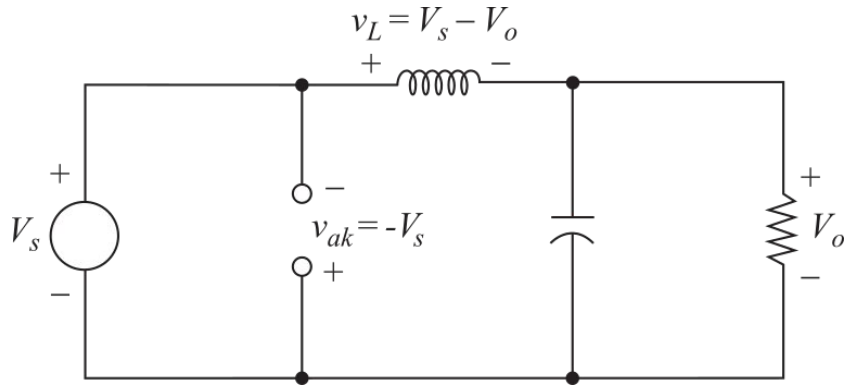
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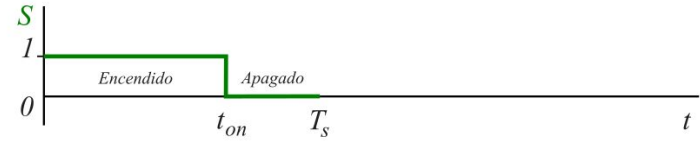
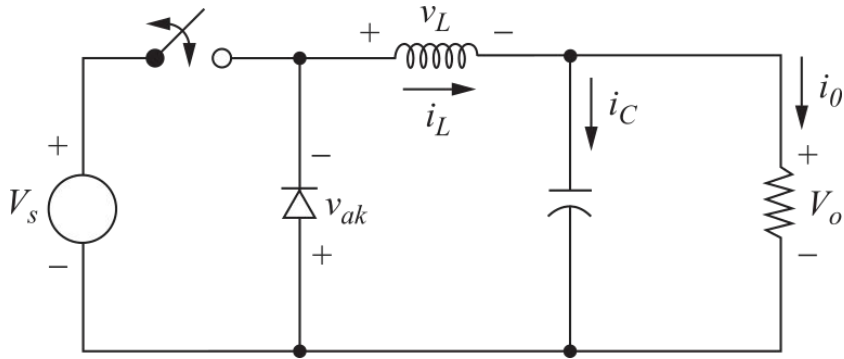
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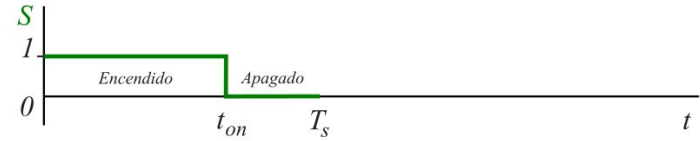
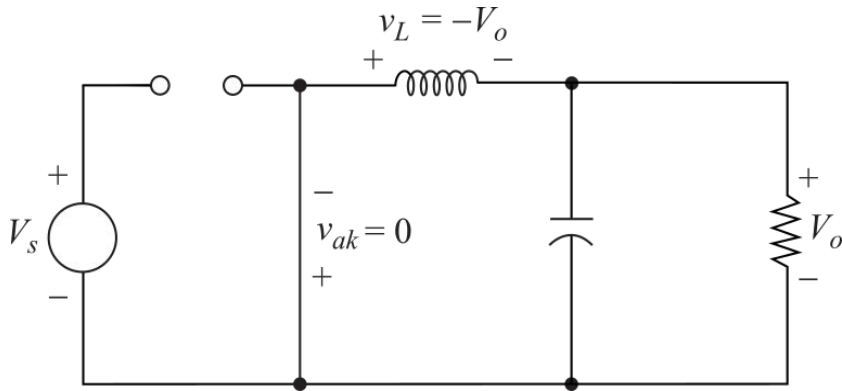
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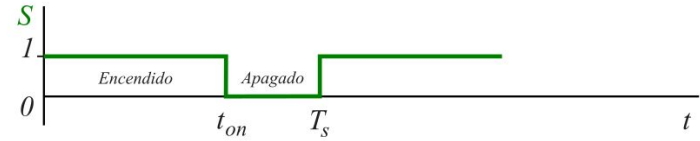
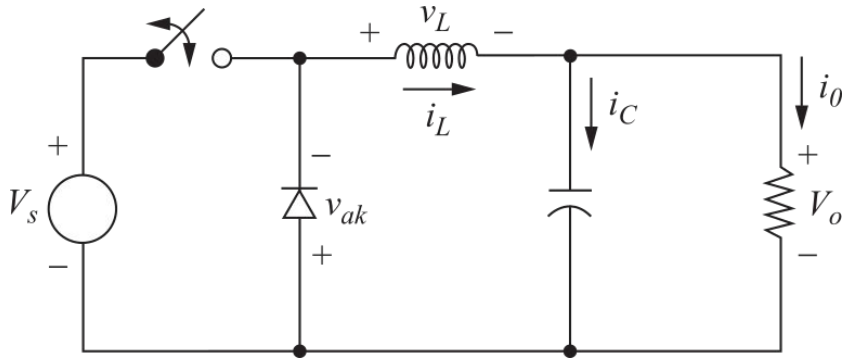
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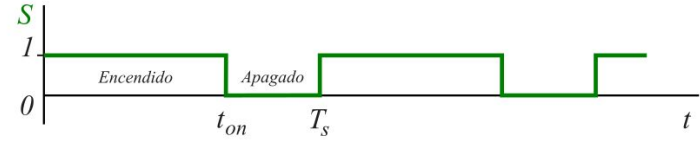
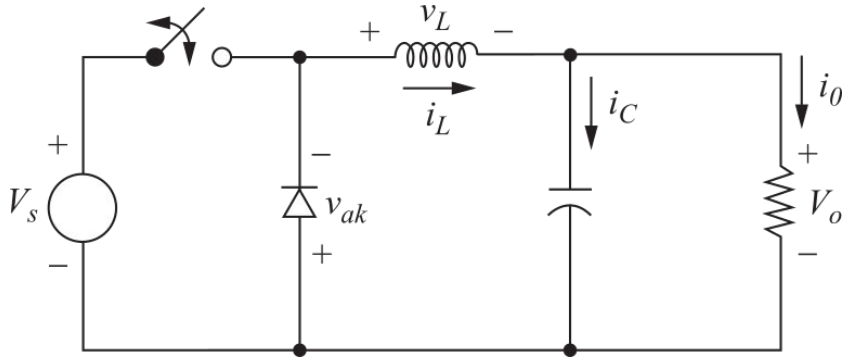
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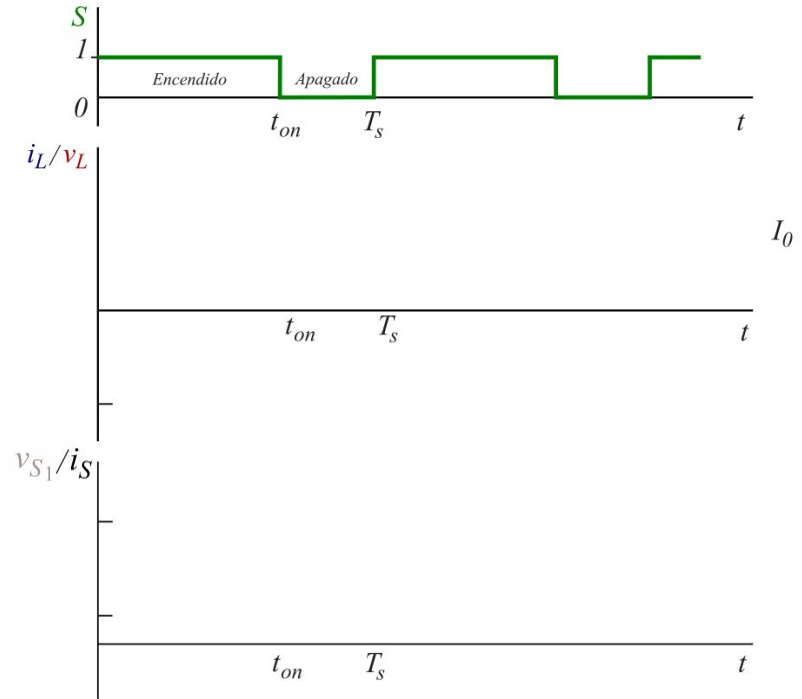
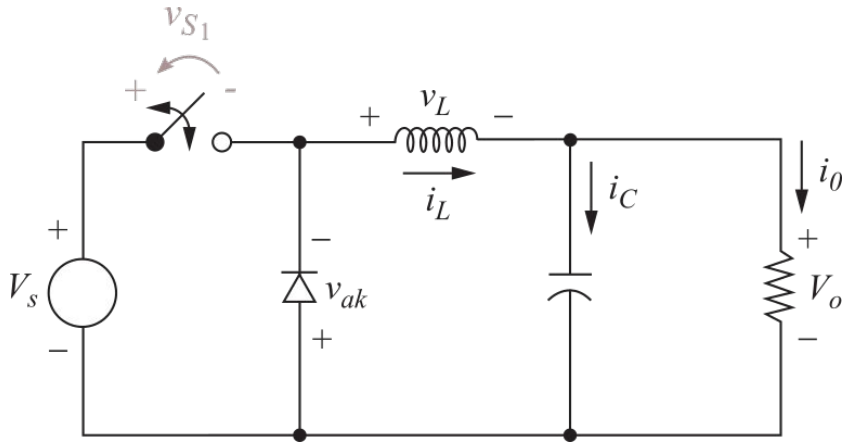
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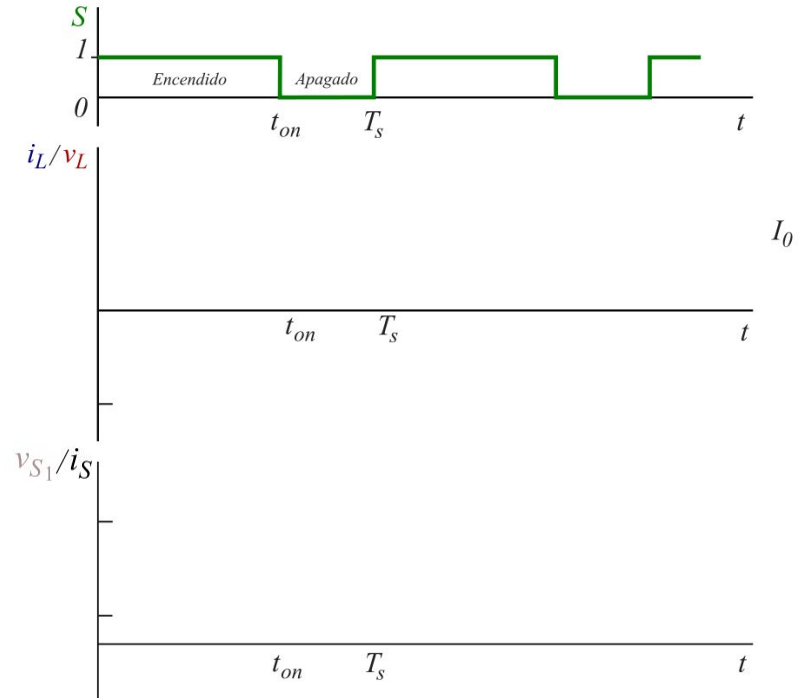
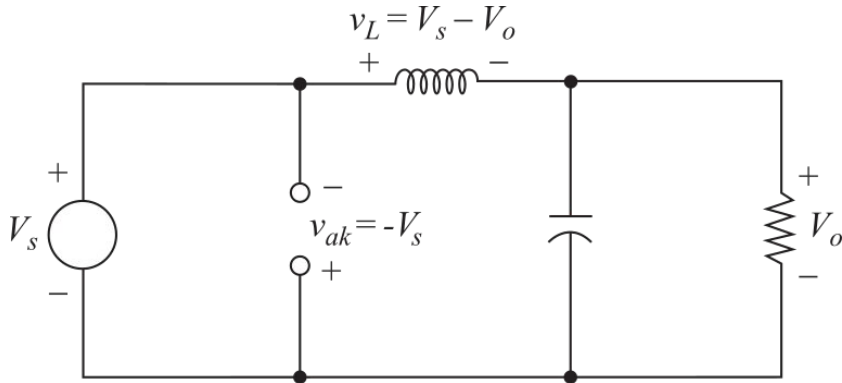
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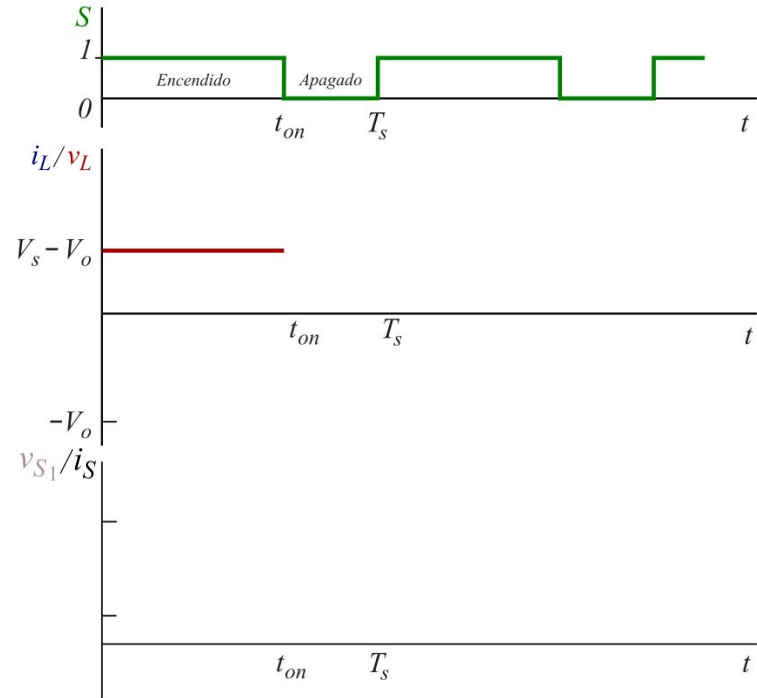
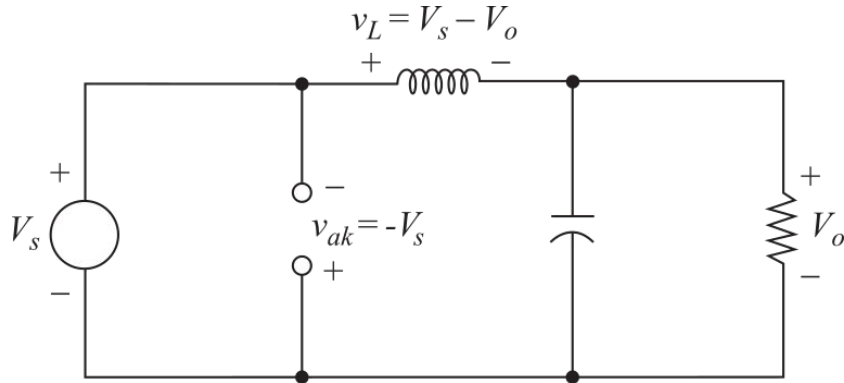
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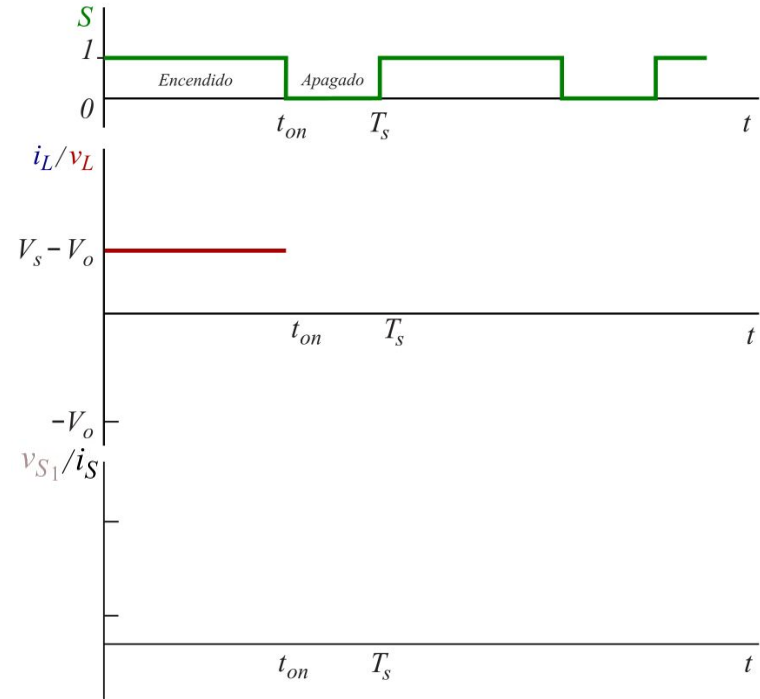
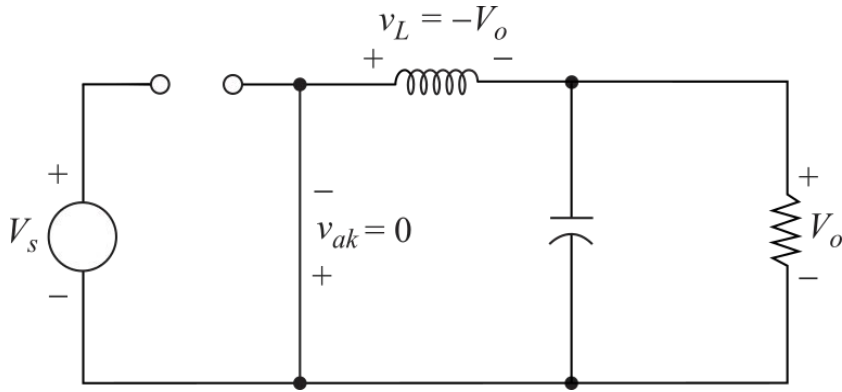
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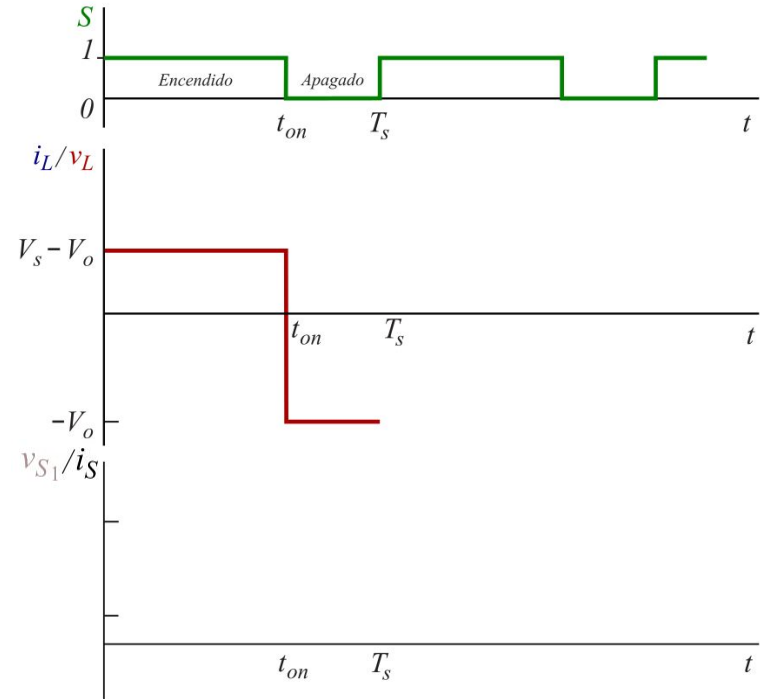
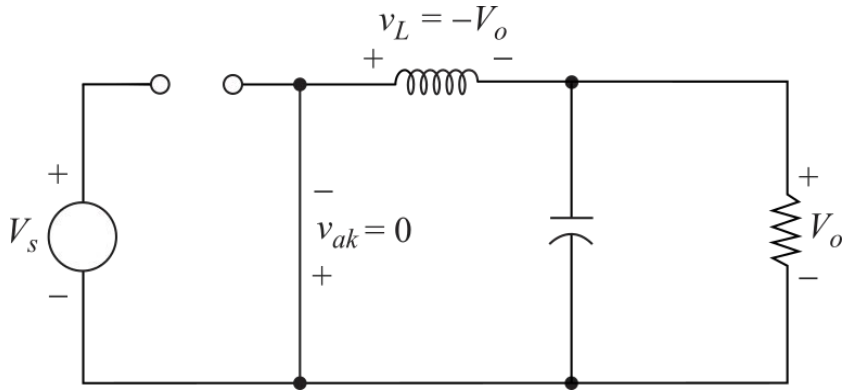
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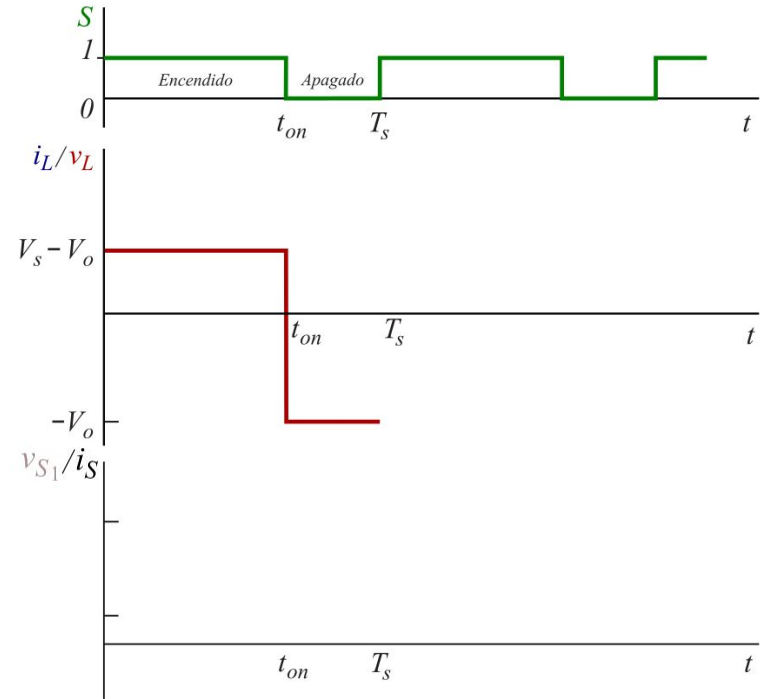
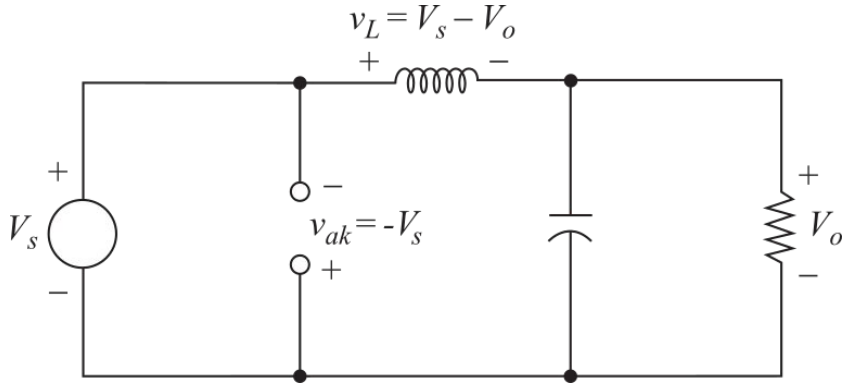
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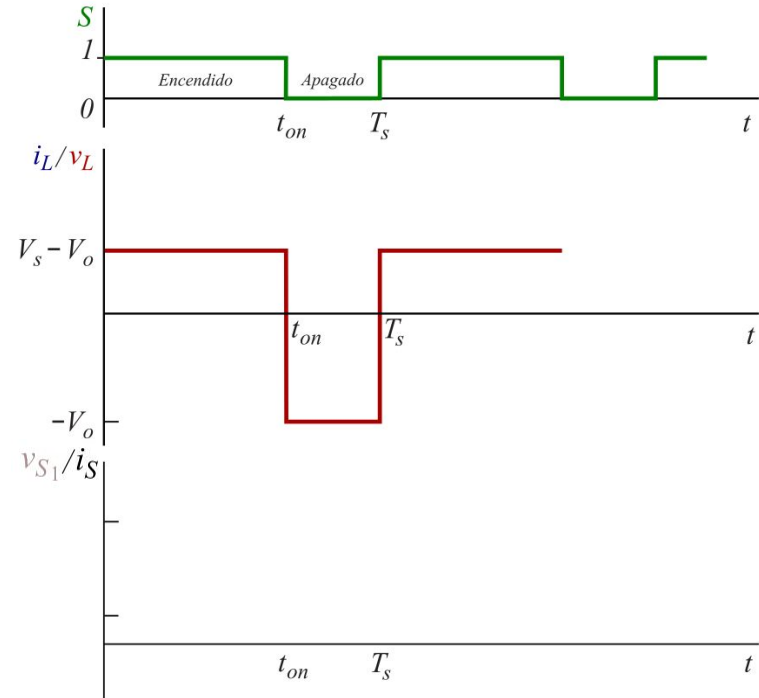
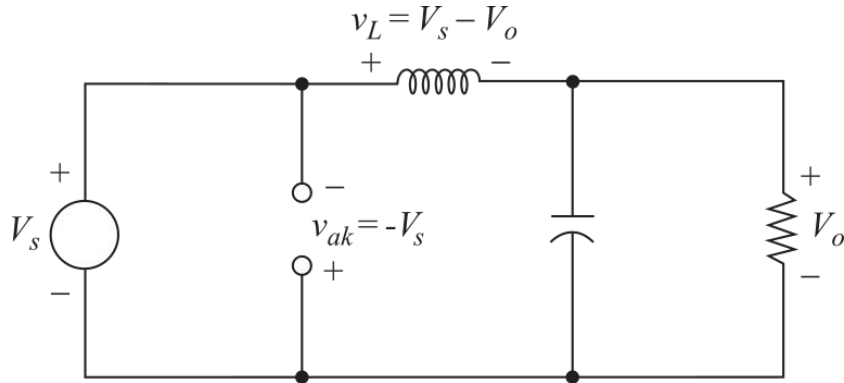
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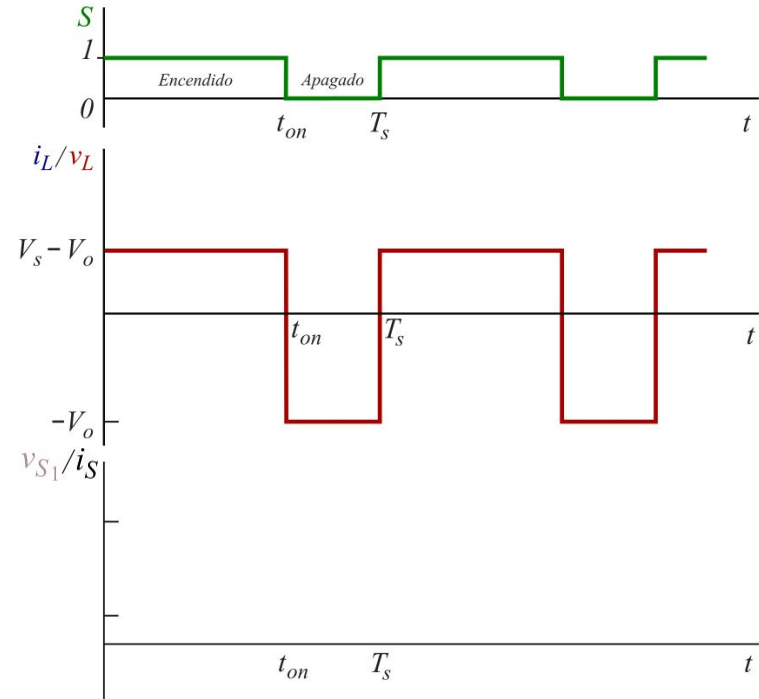
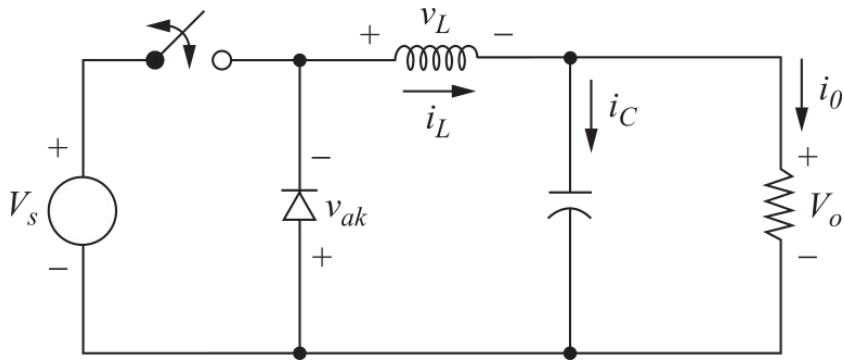
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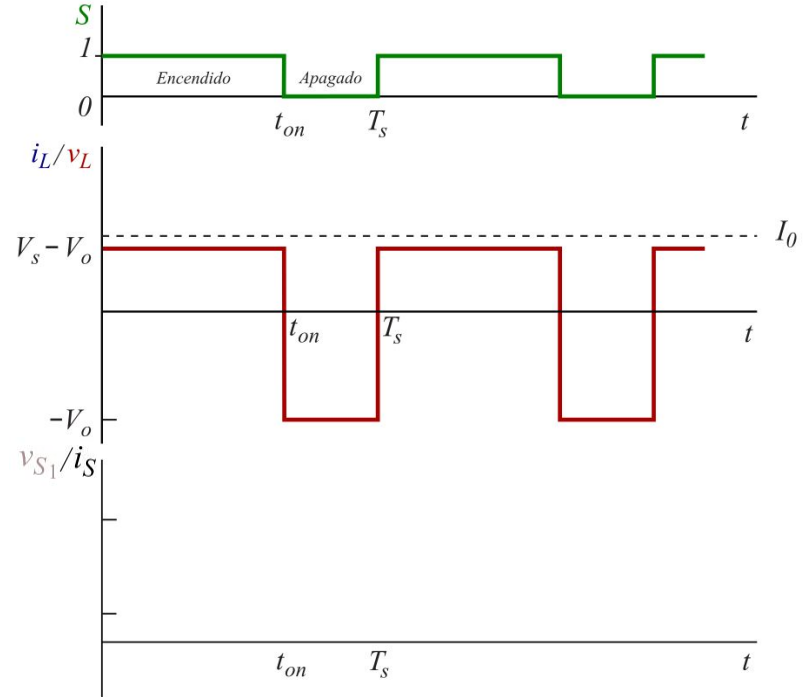
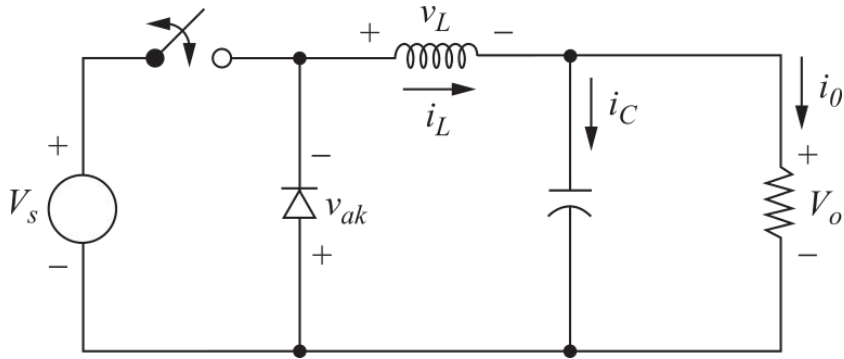
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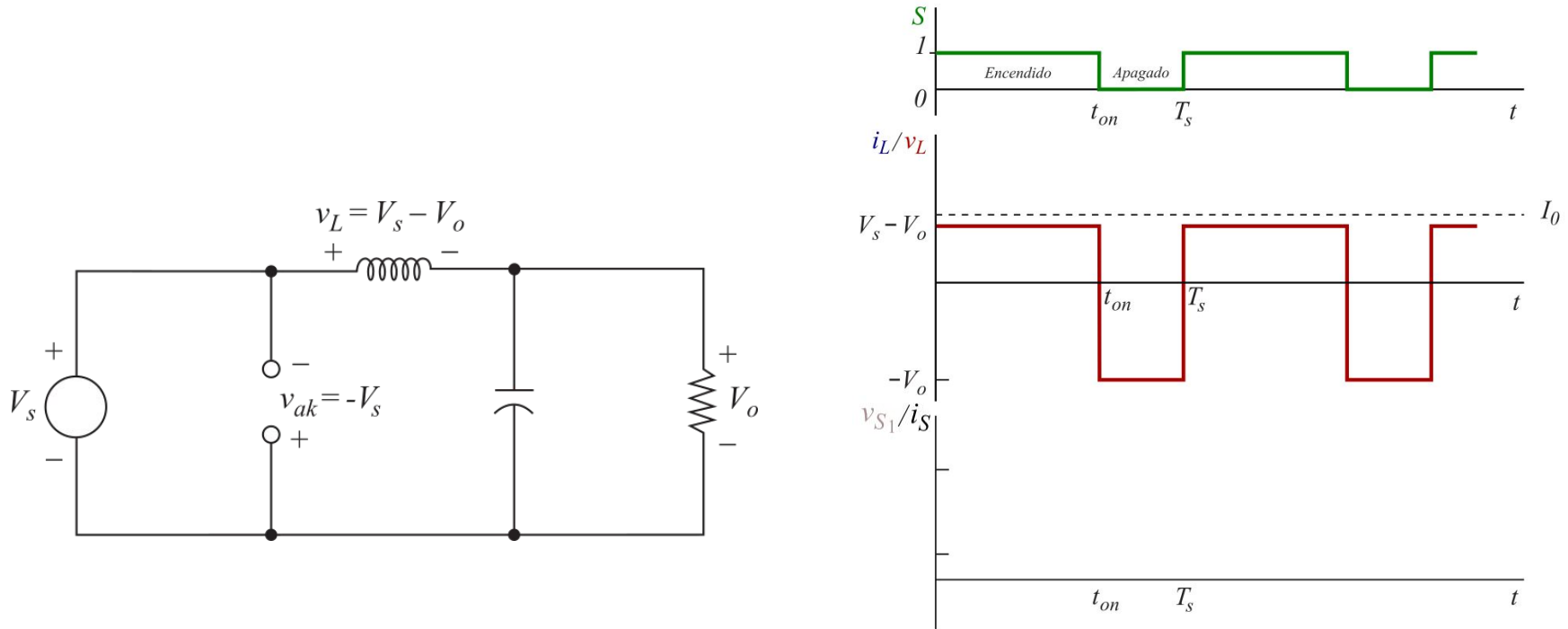
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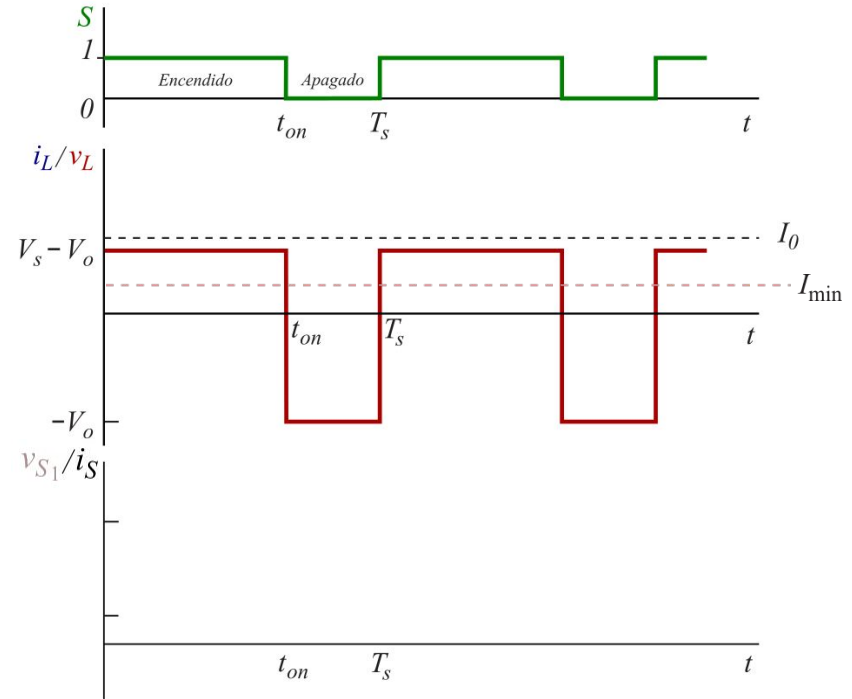
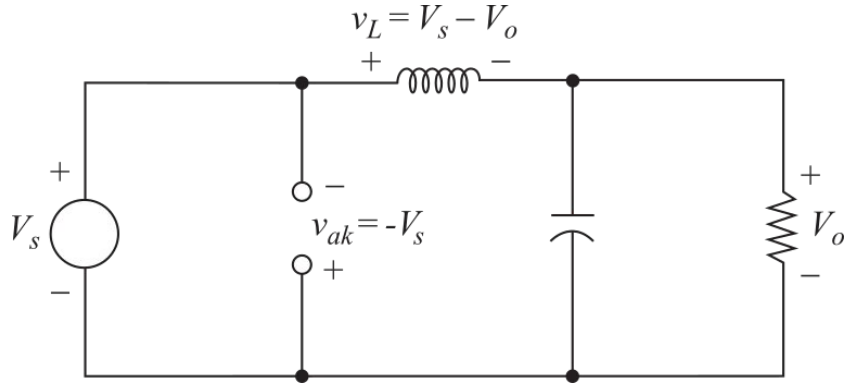
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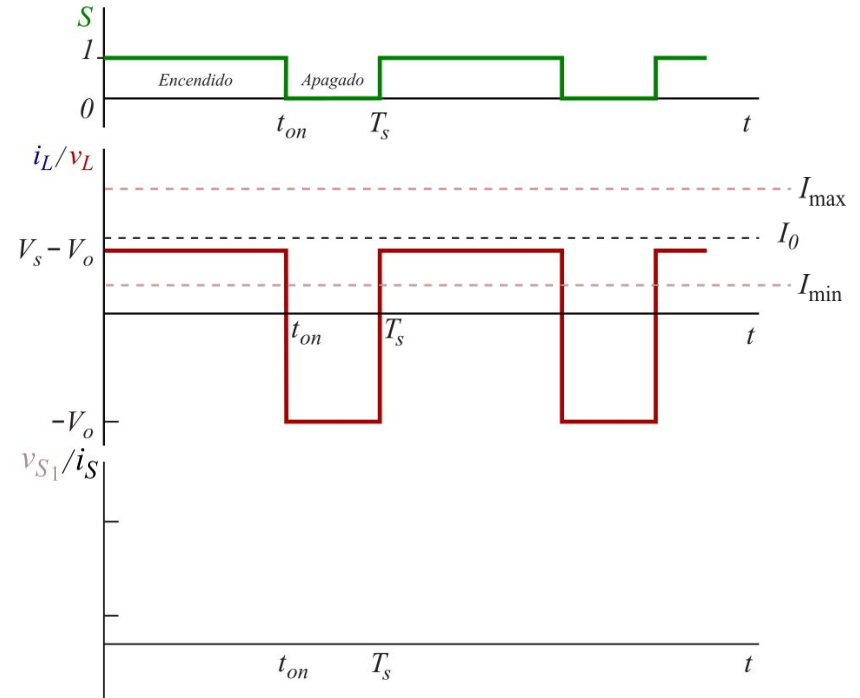
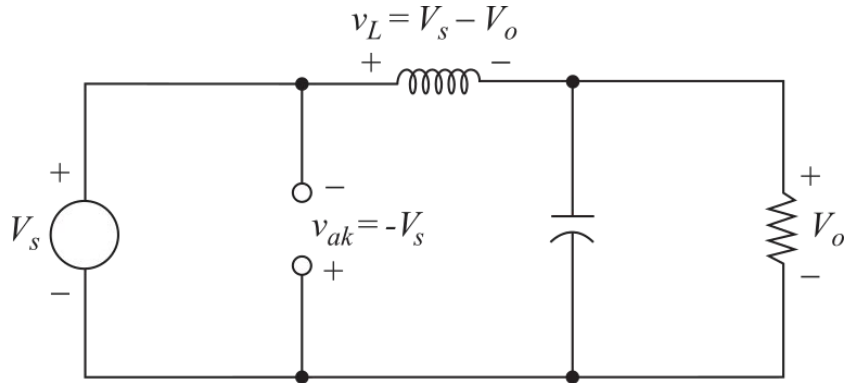
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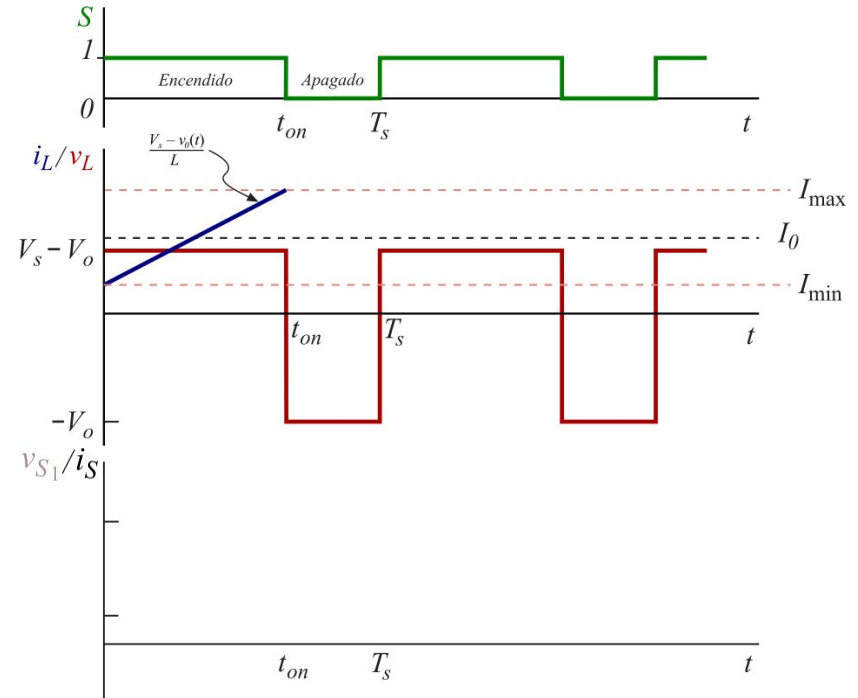
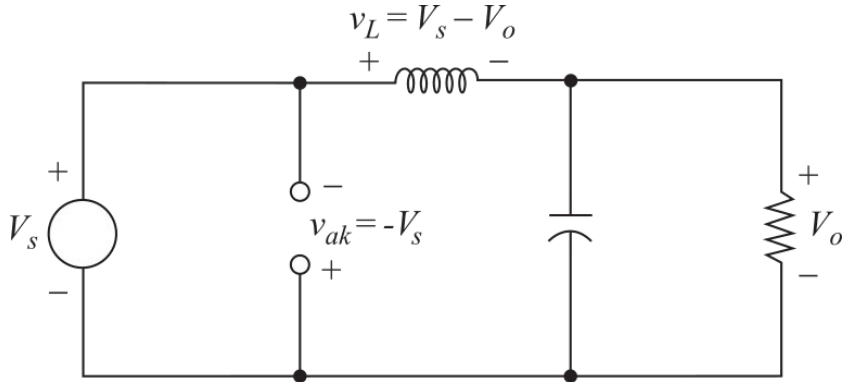
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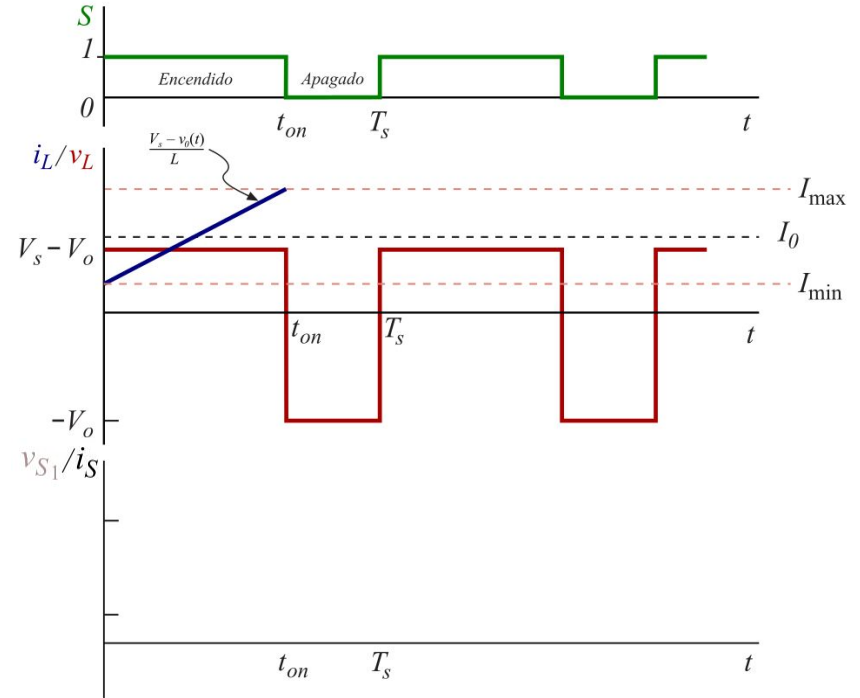
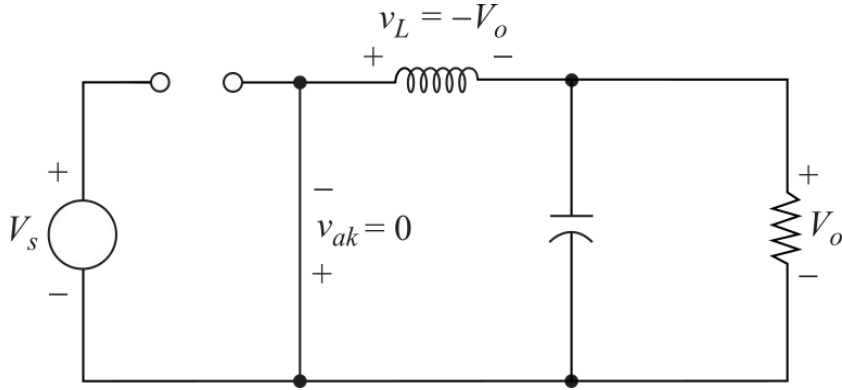
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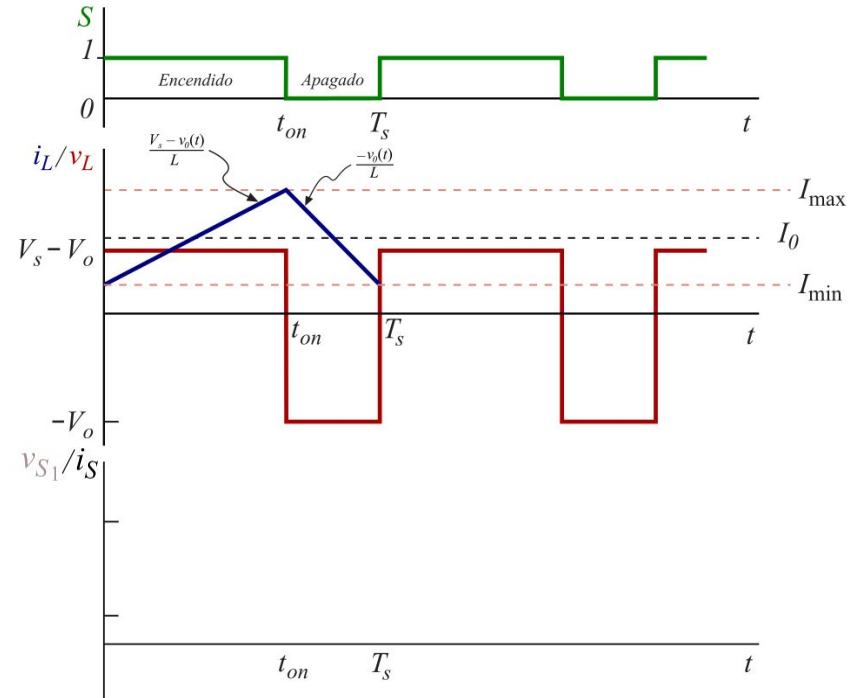
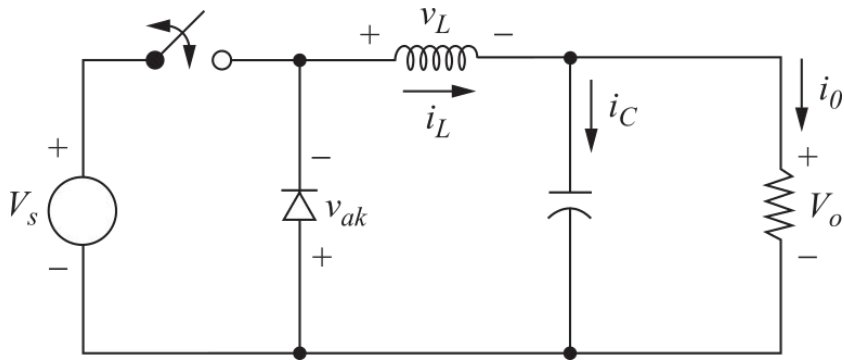
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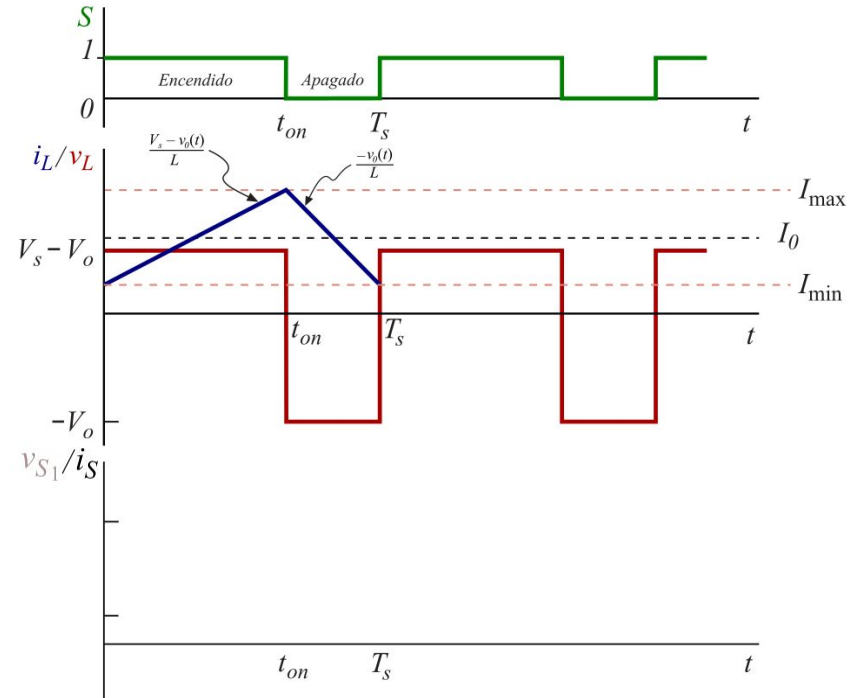
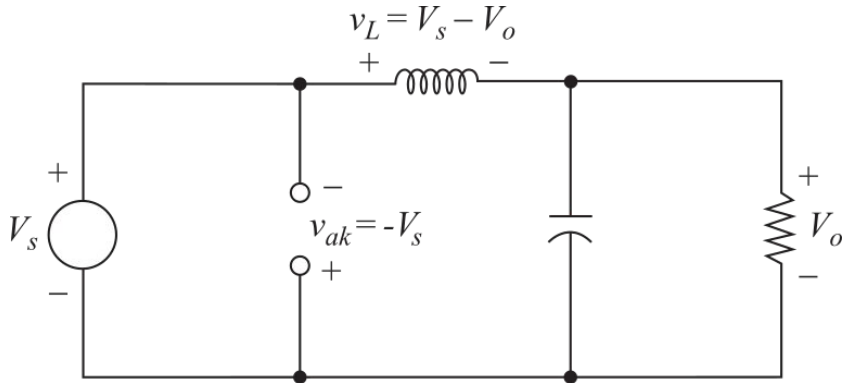
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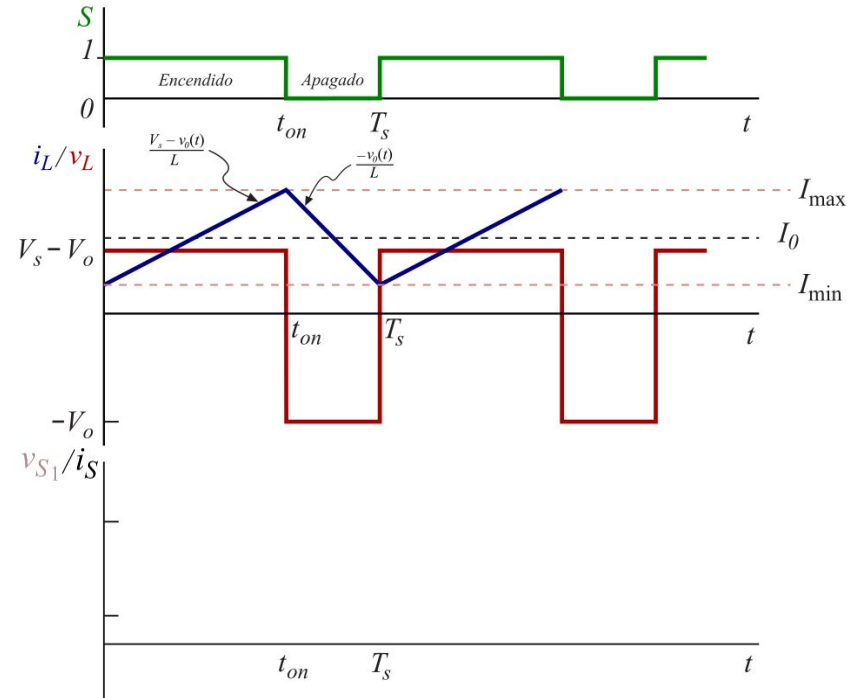
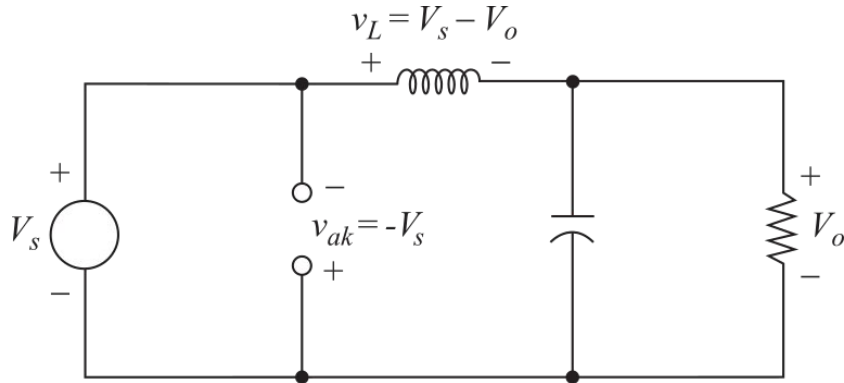
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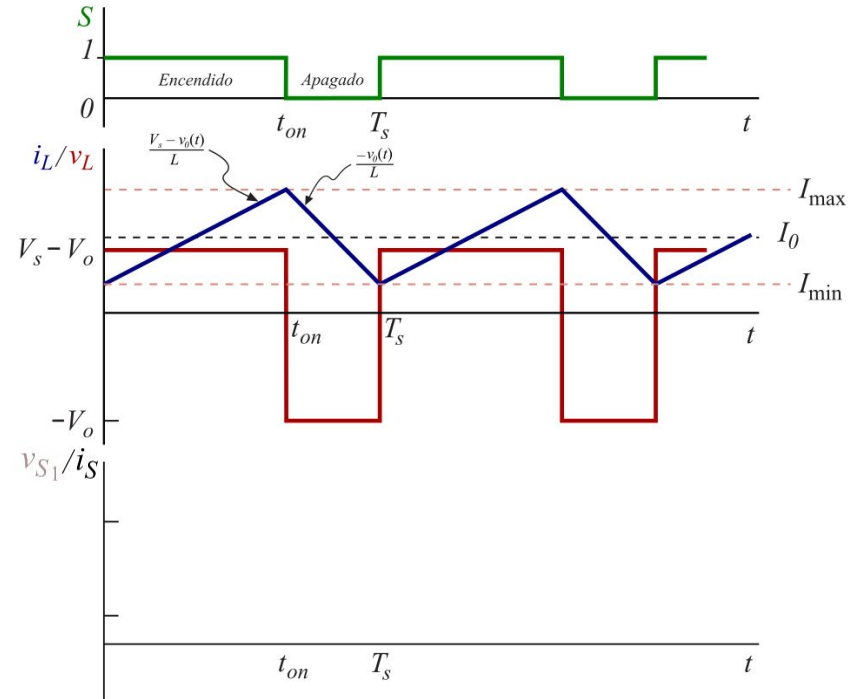
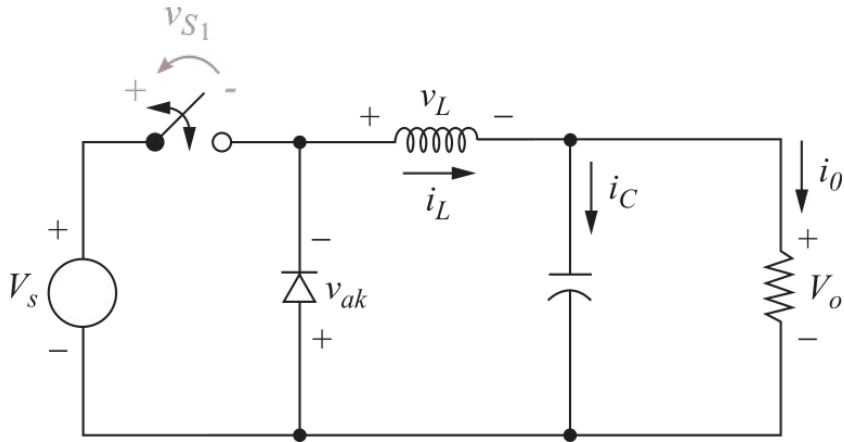
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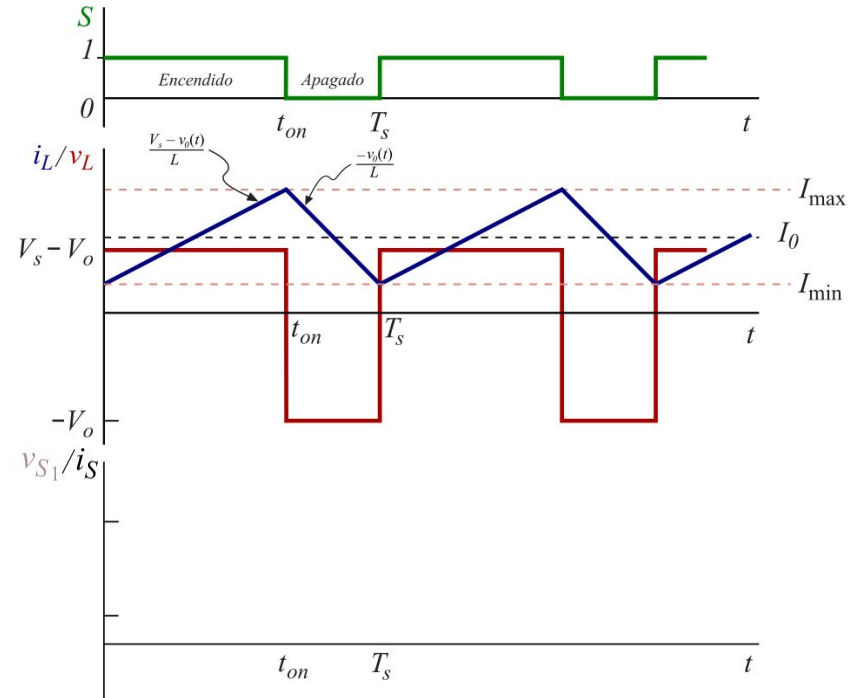
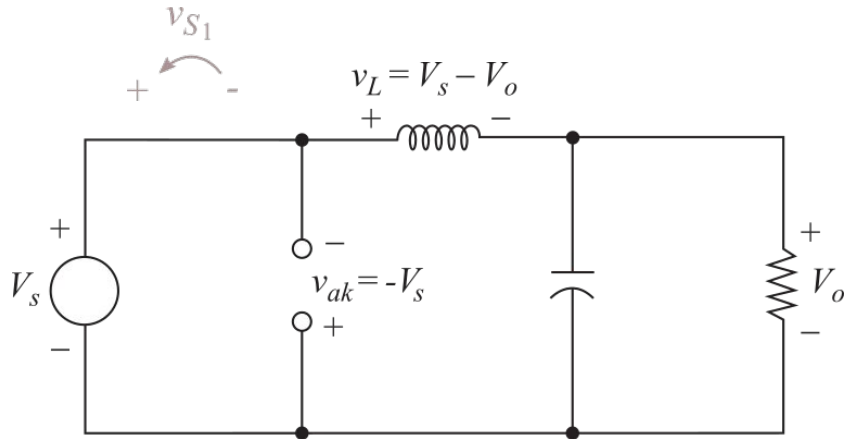
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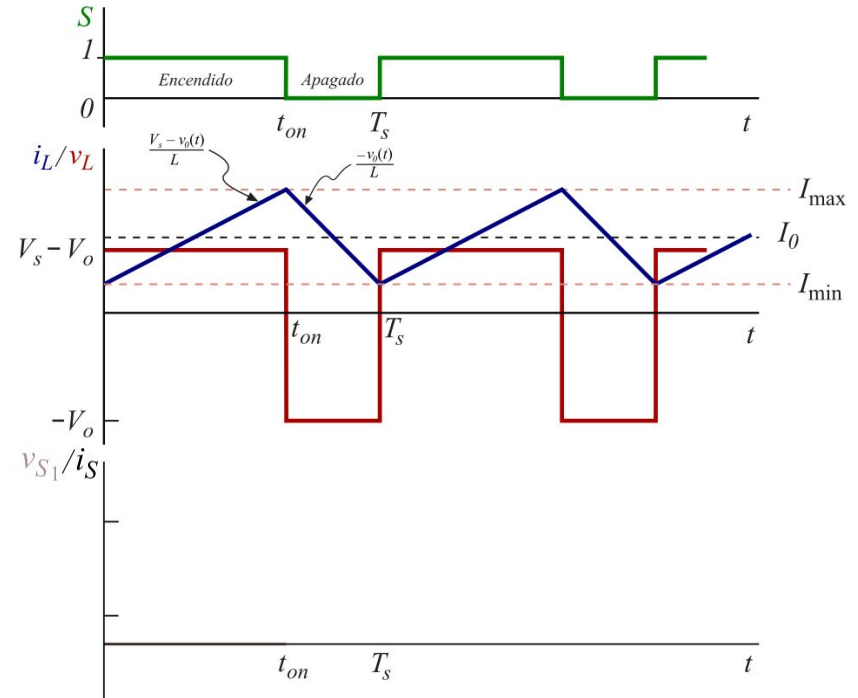
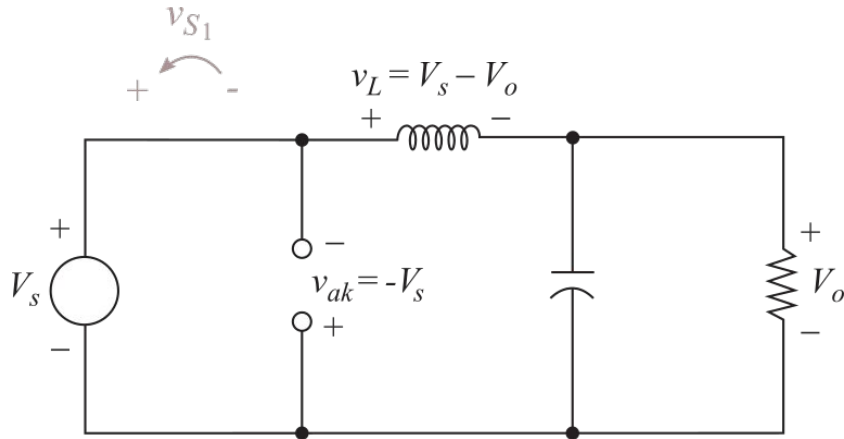
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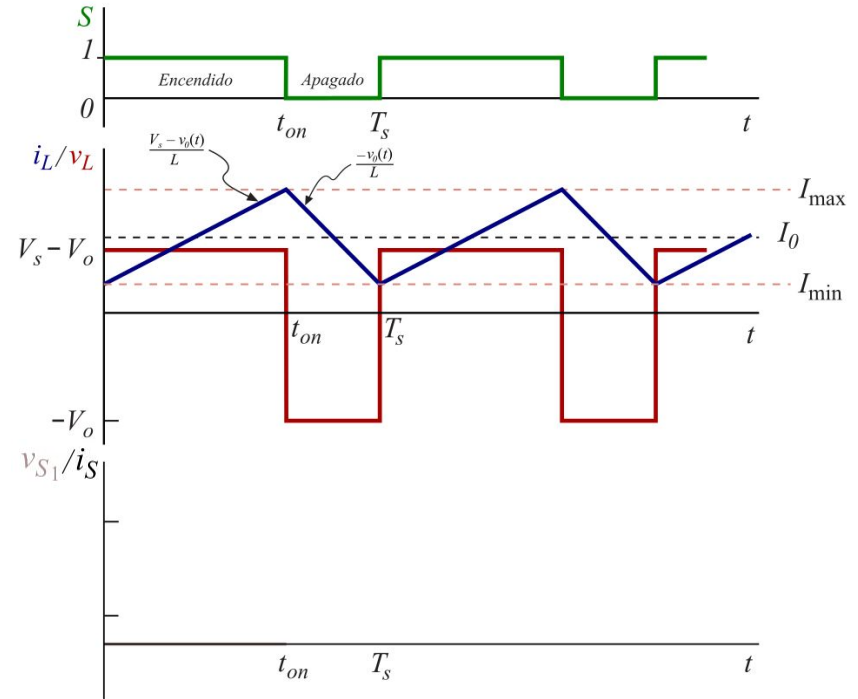
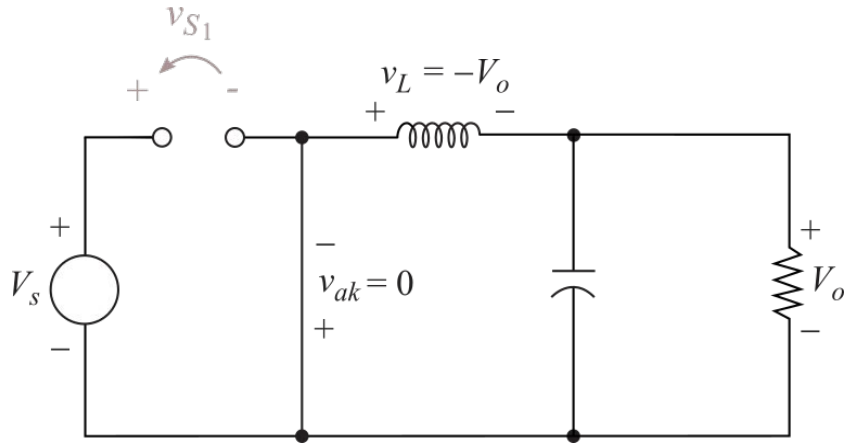
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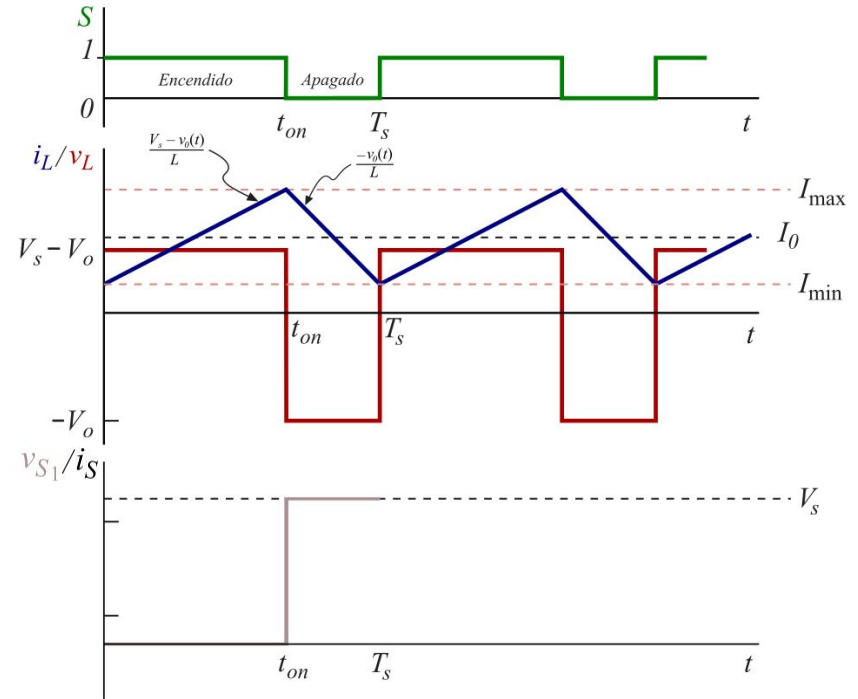
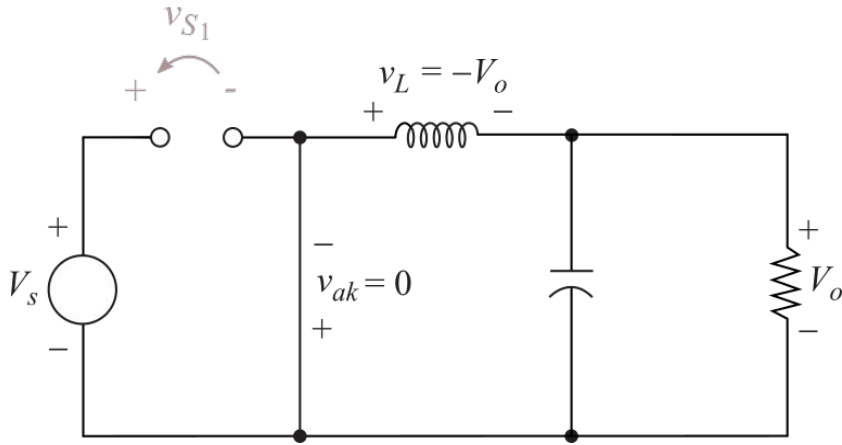
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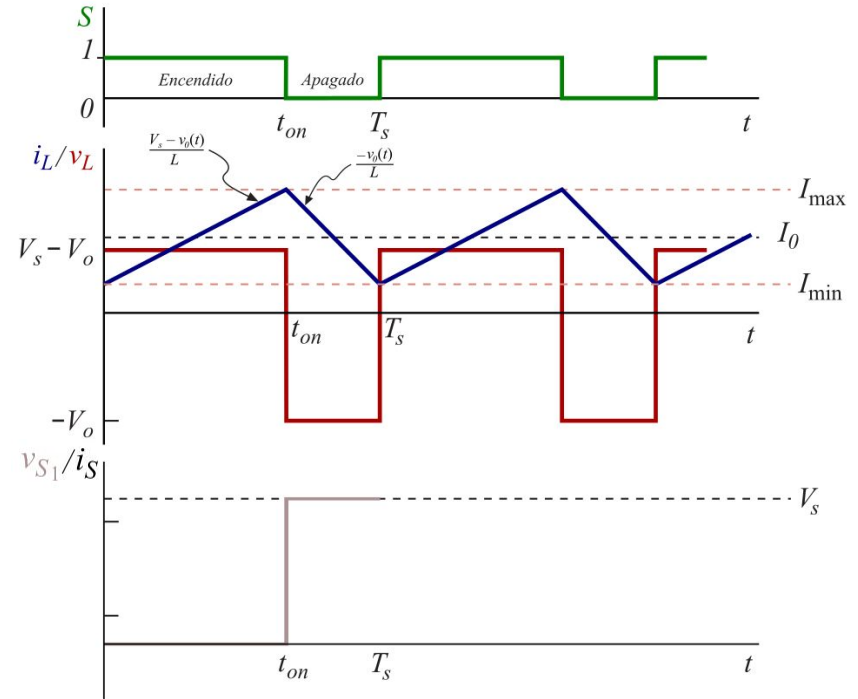
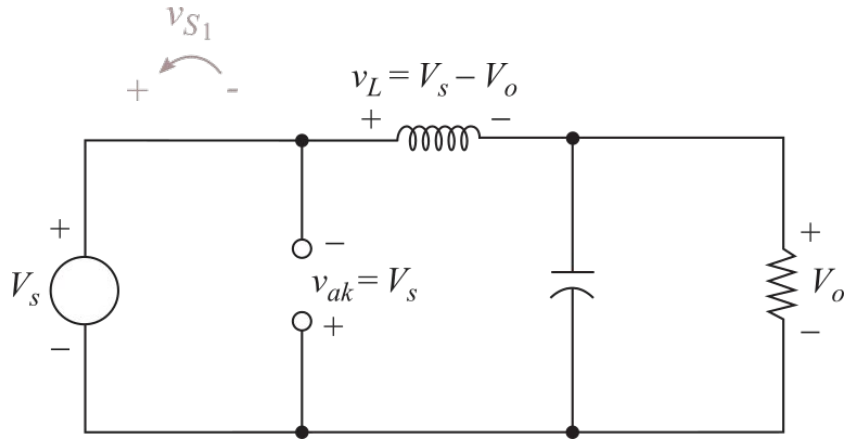
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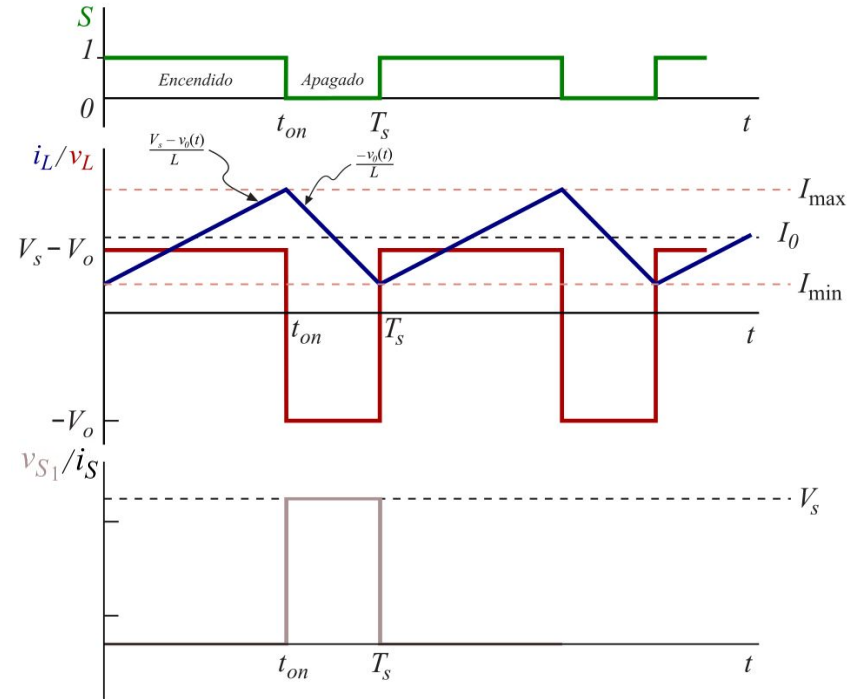
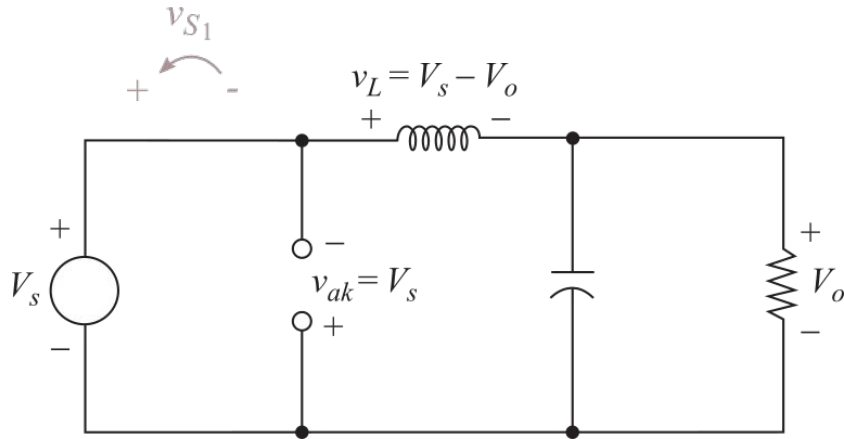
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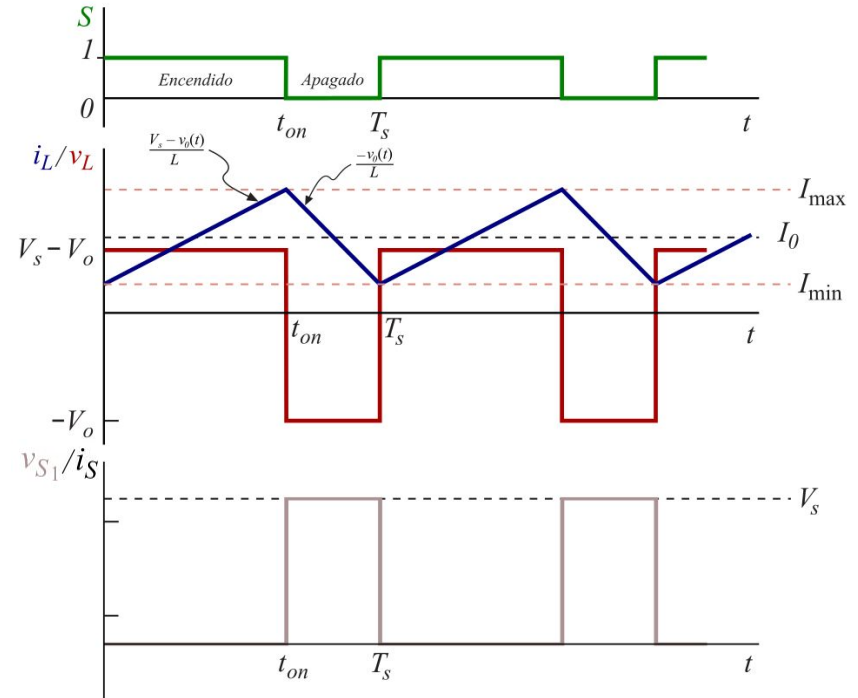
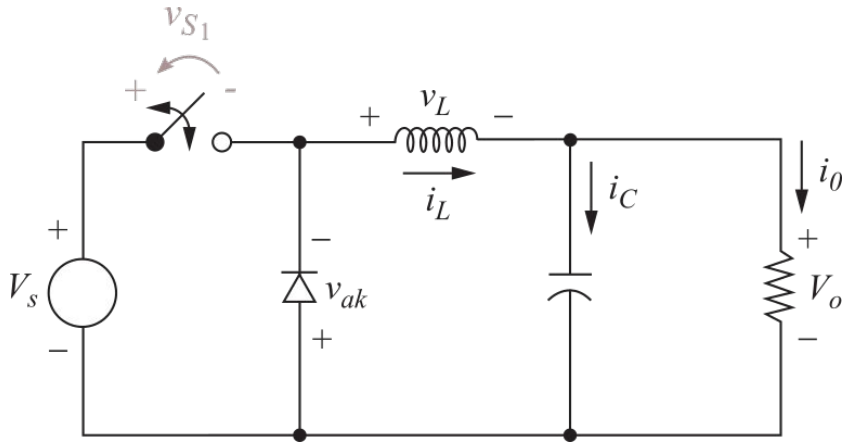
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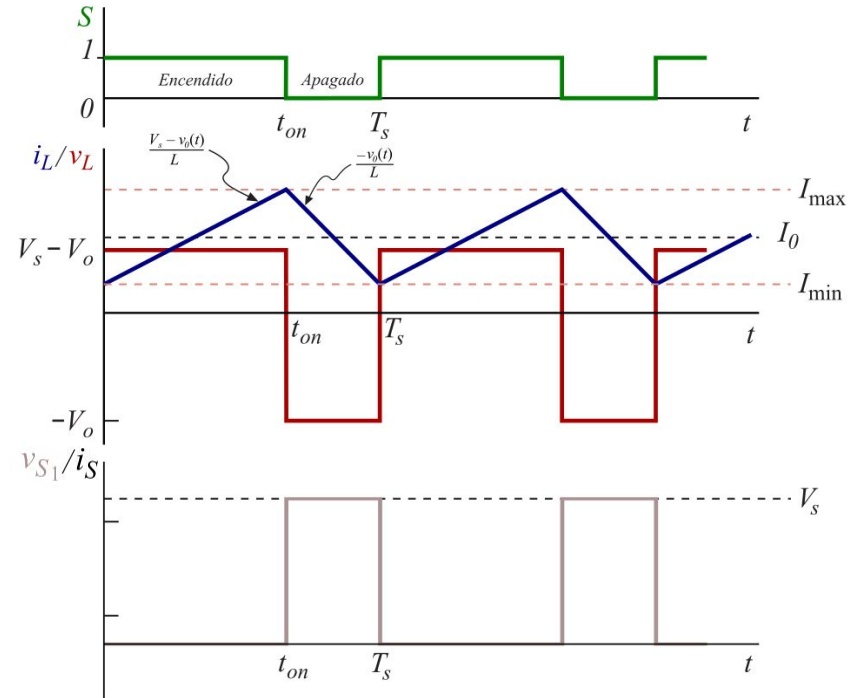
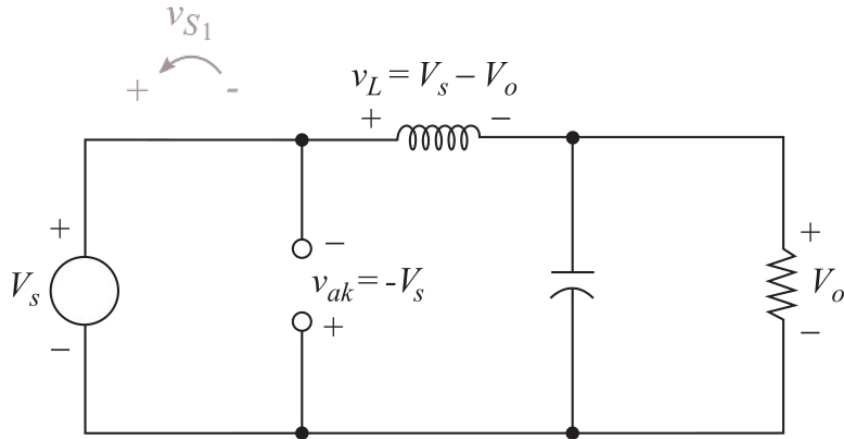
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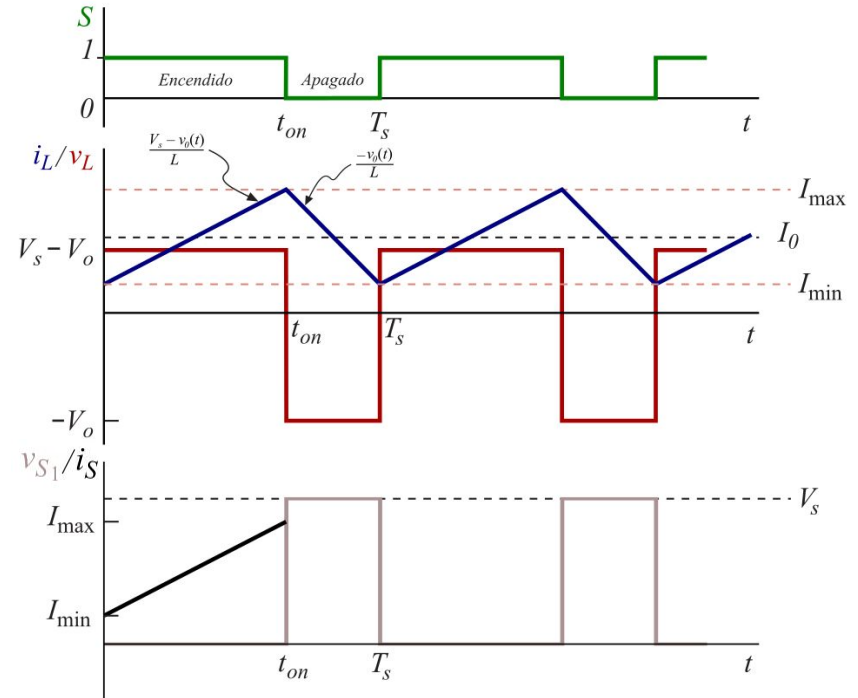
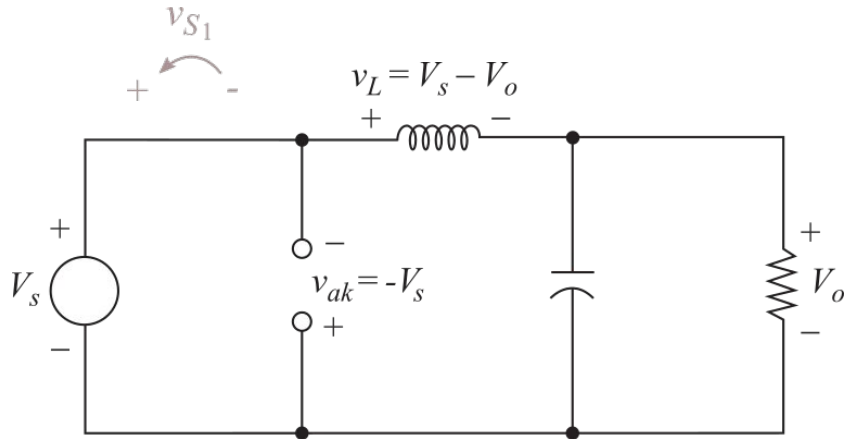
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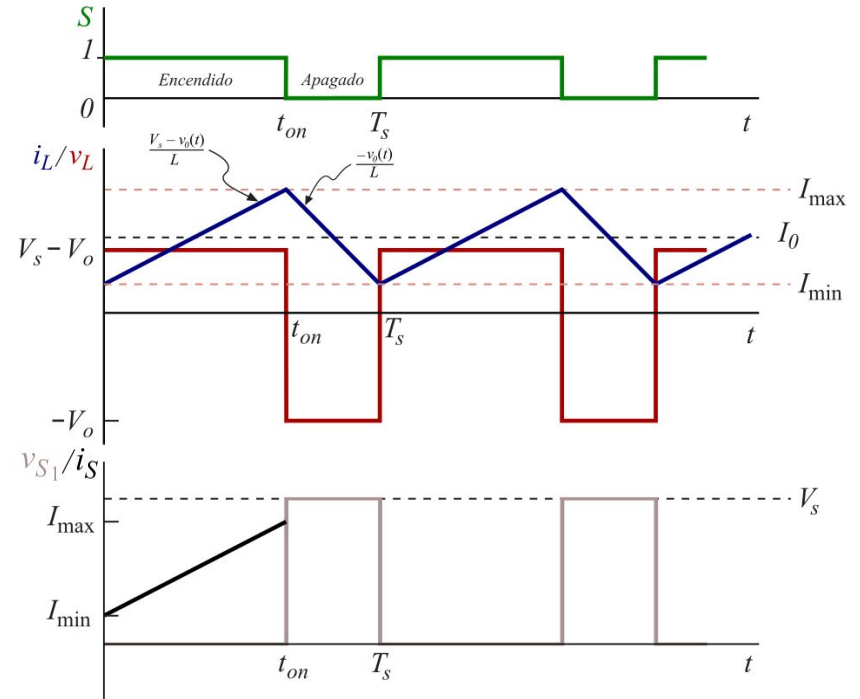
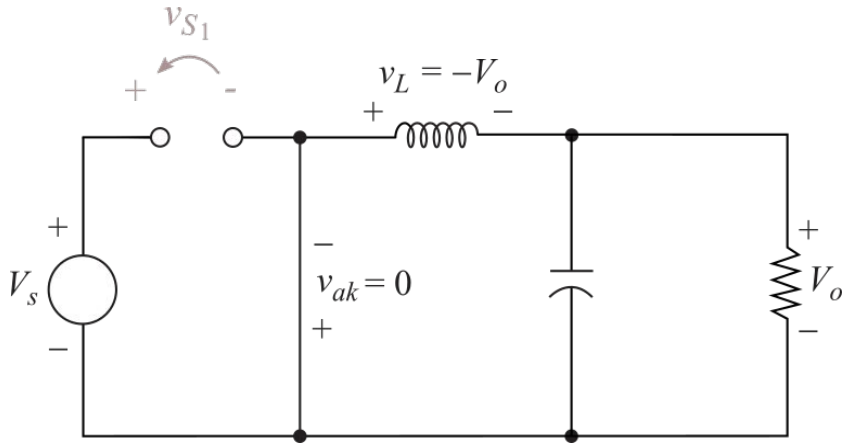
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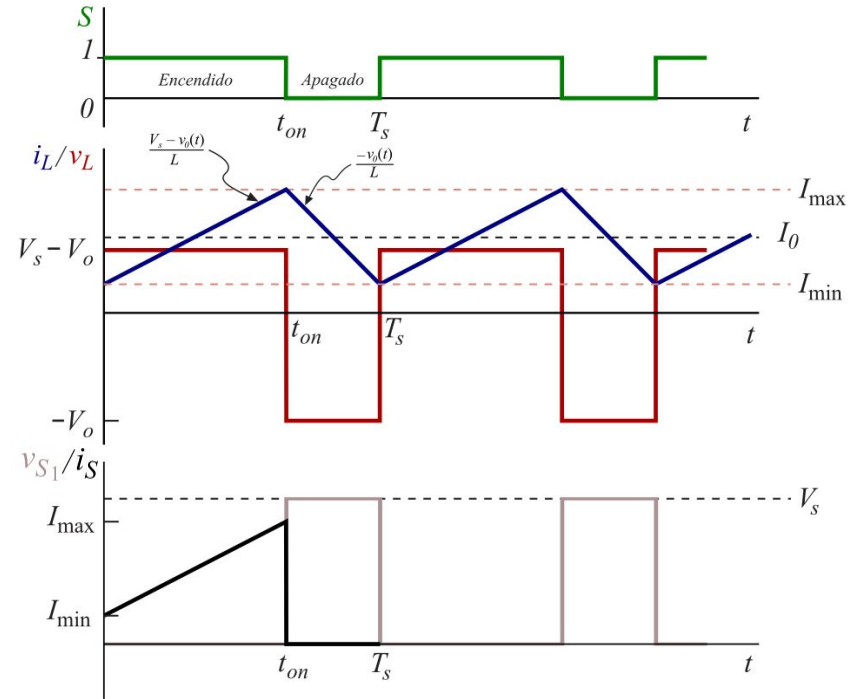
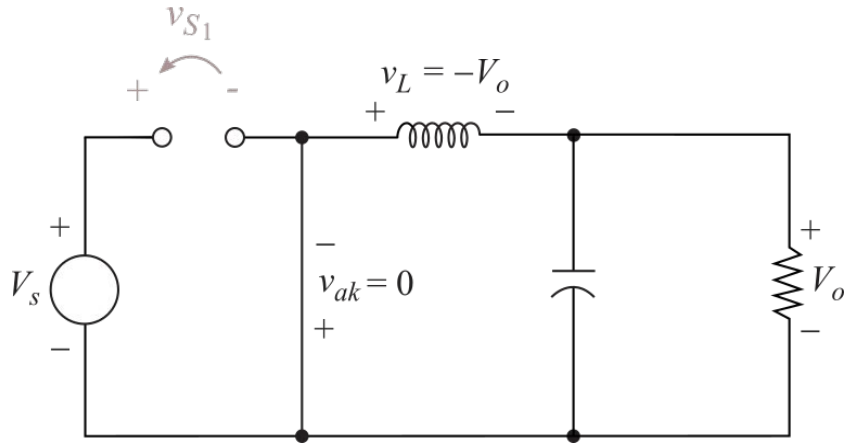
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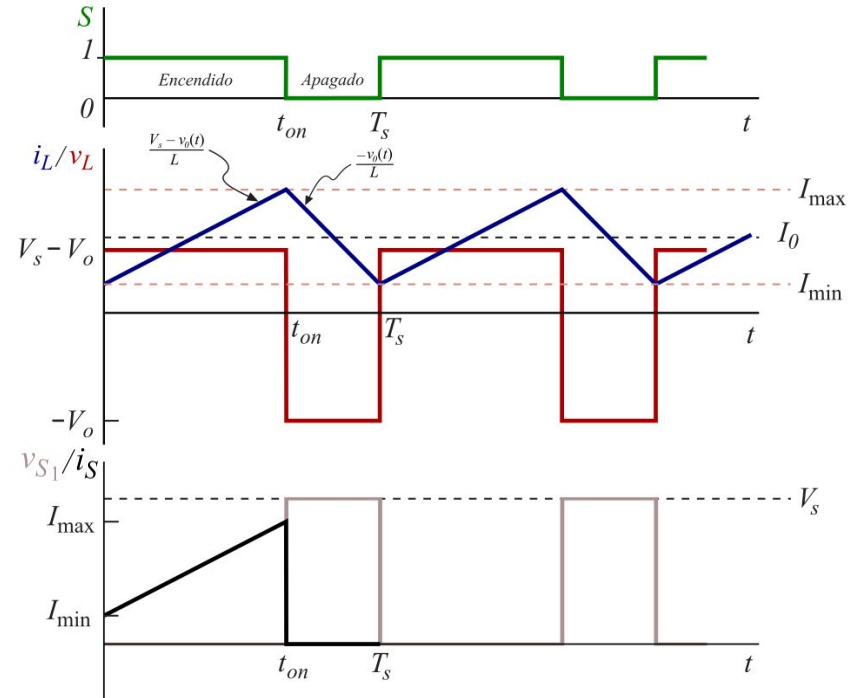
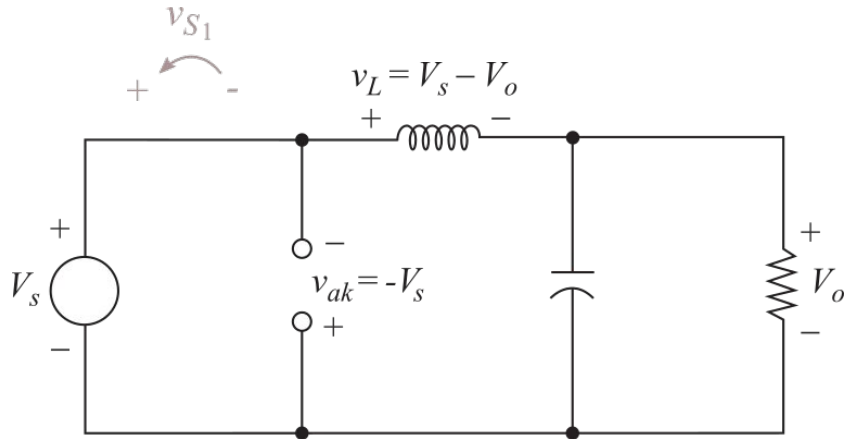
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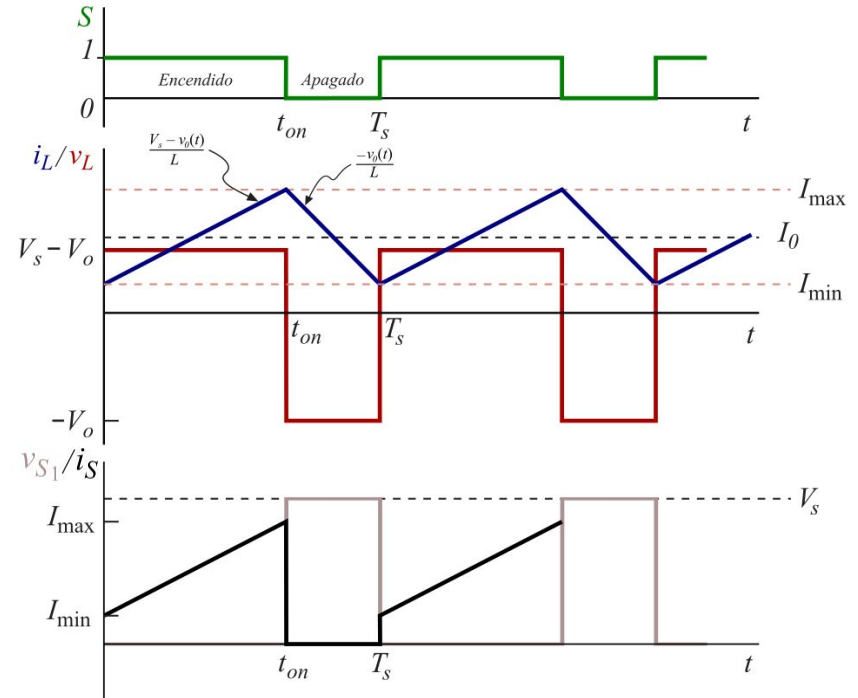
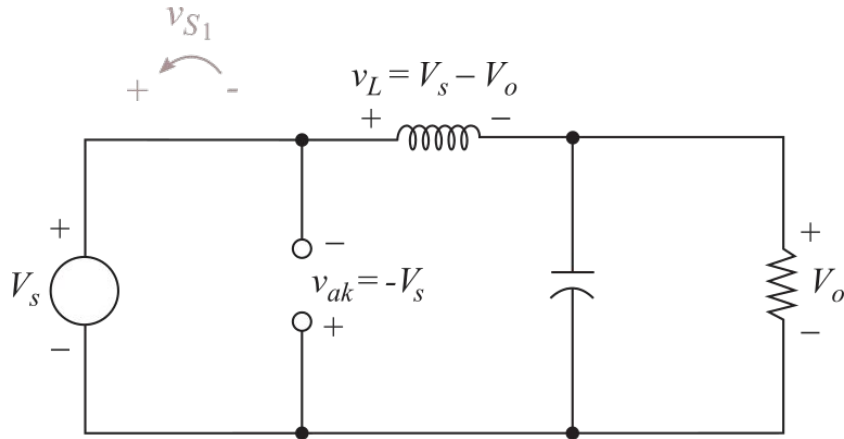
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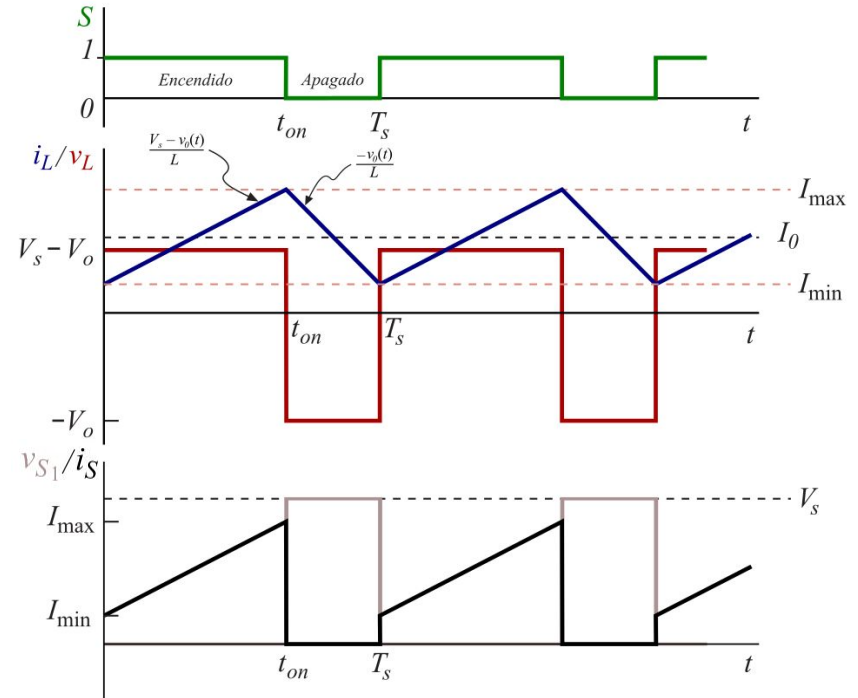
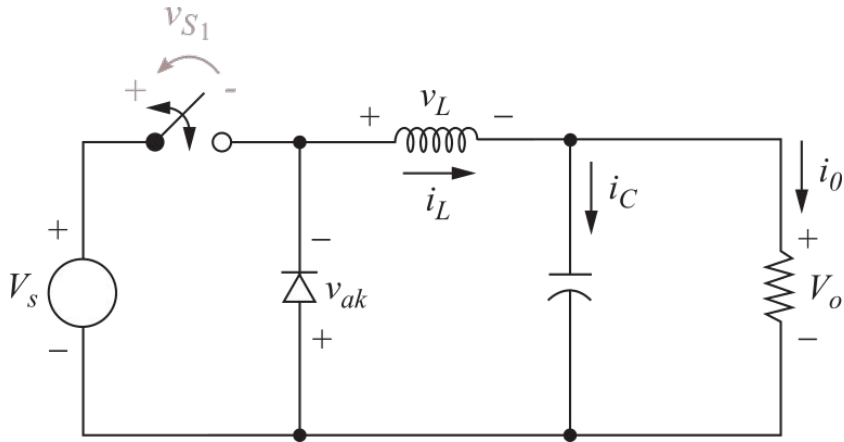
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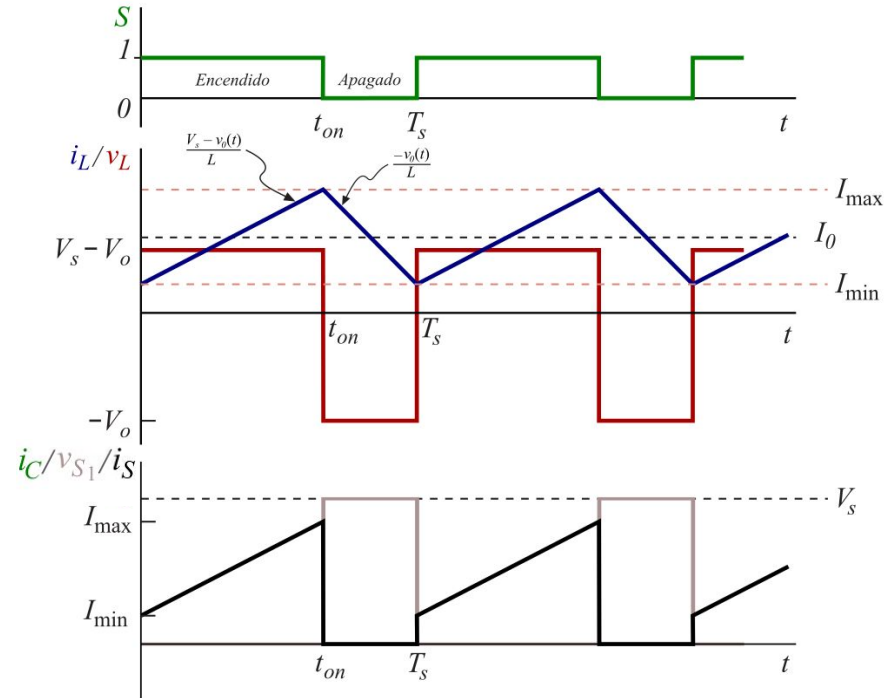
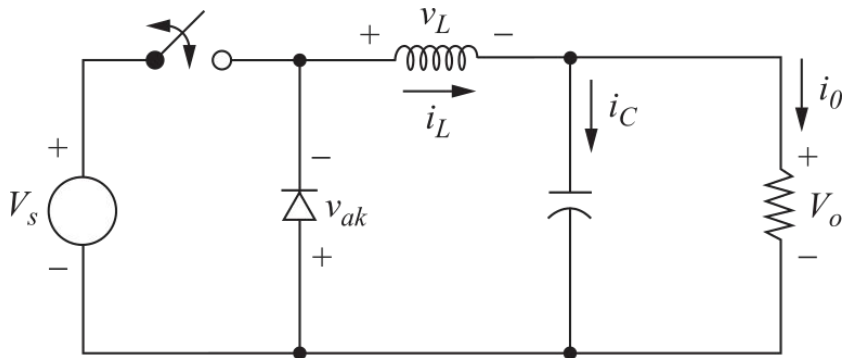
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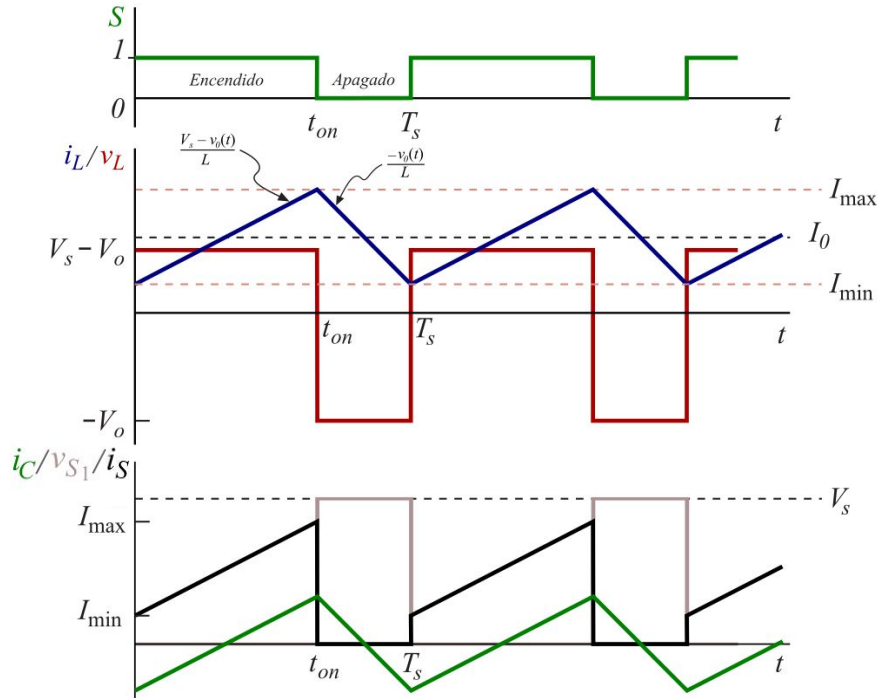
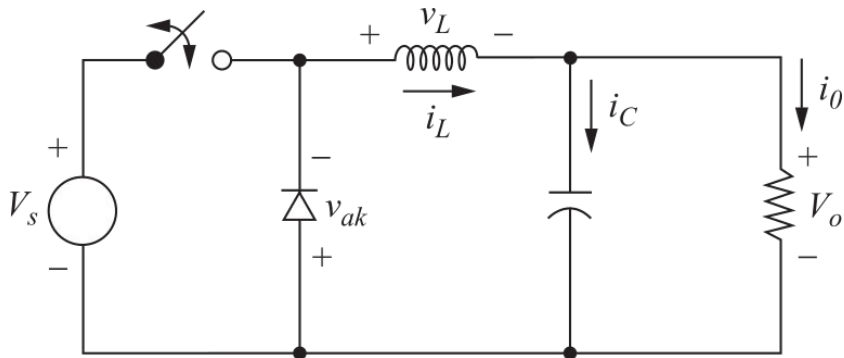
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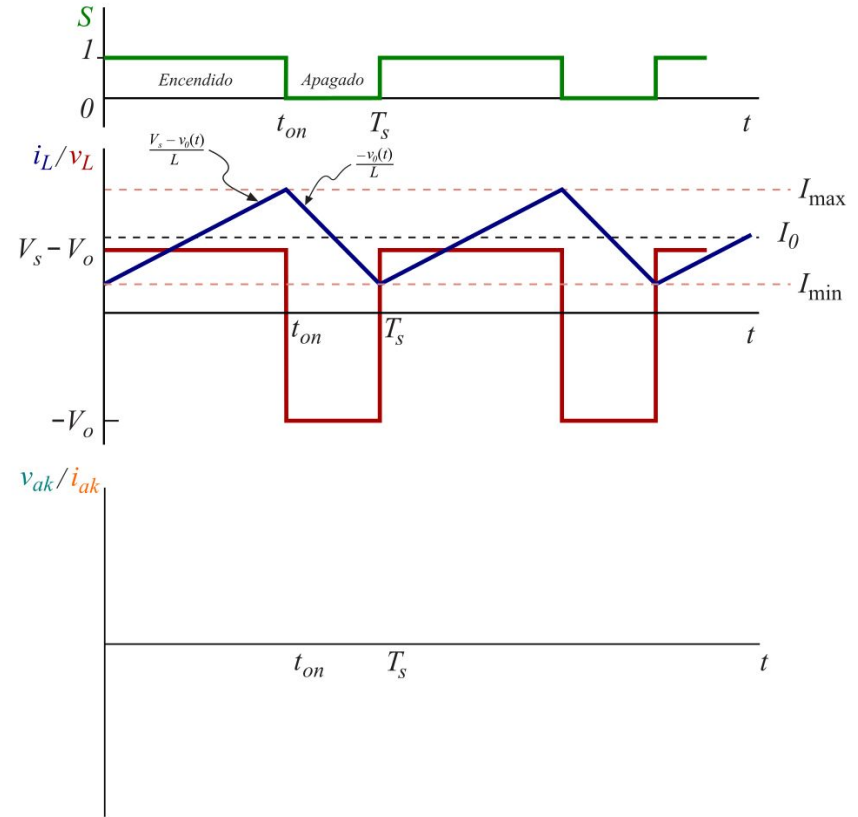
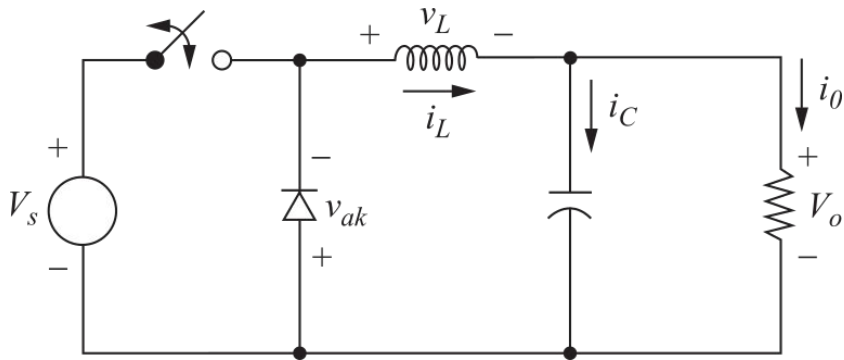
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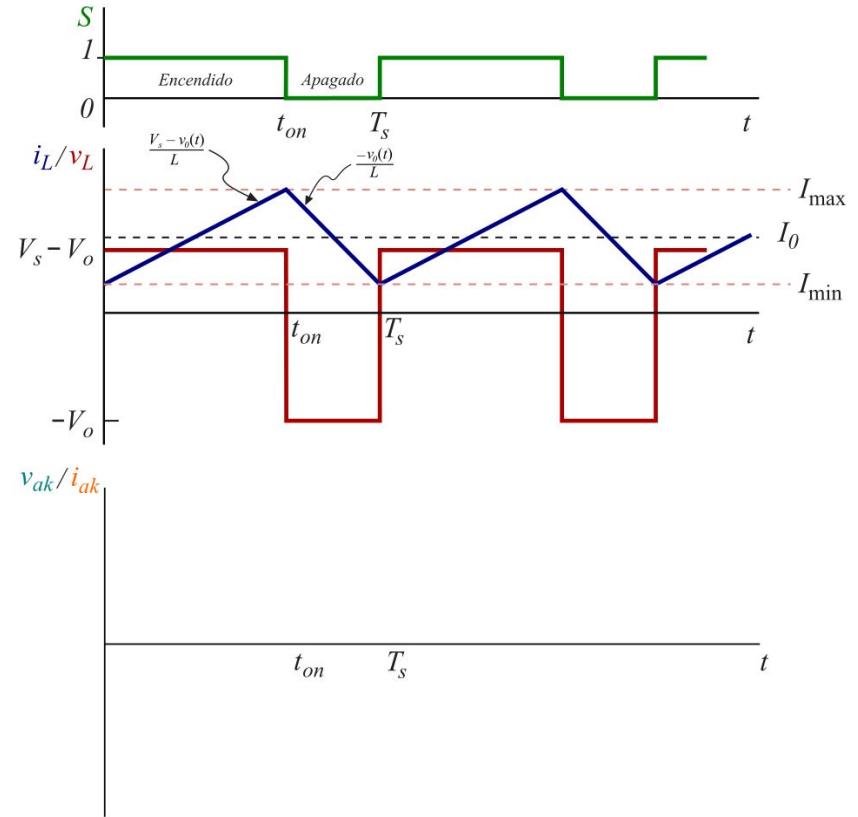
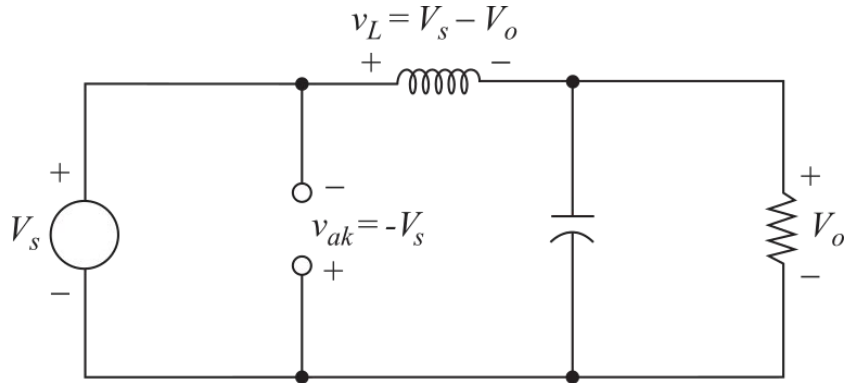
Convertidor Reductor: formas de onda



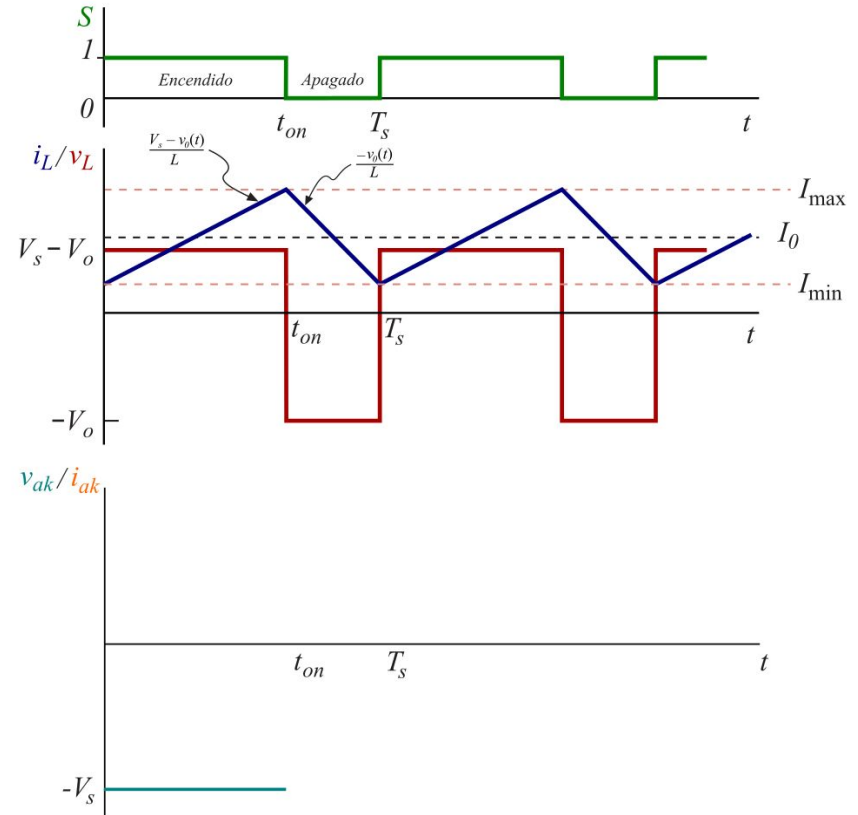
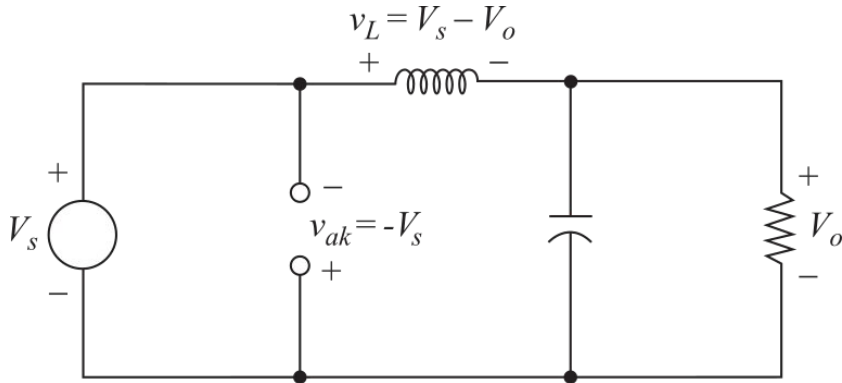
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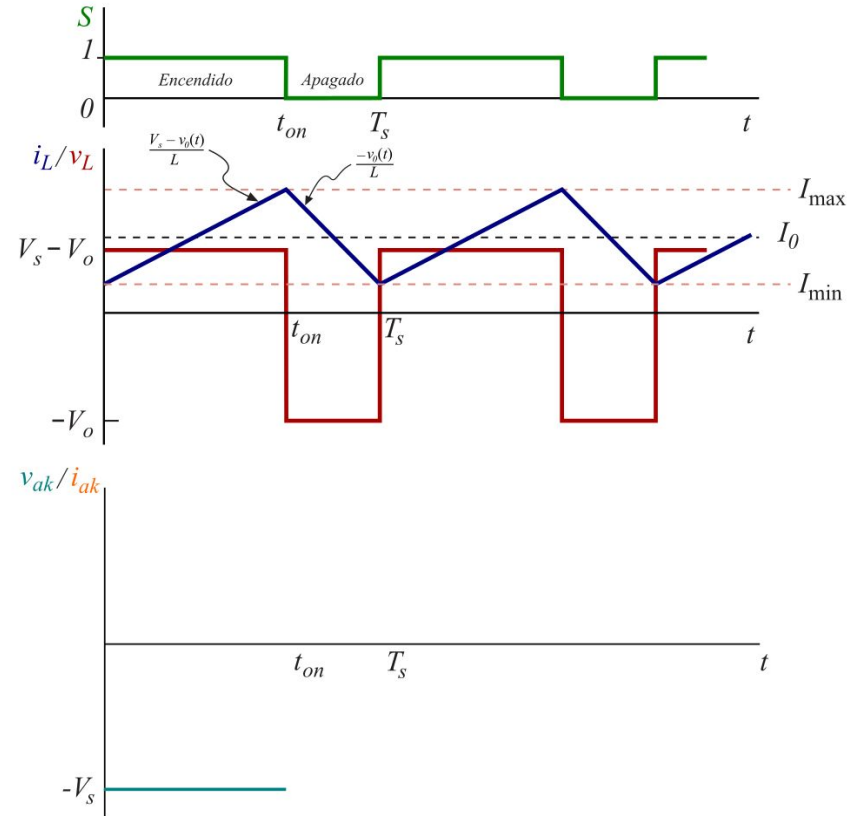
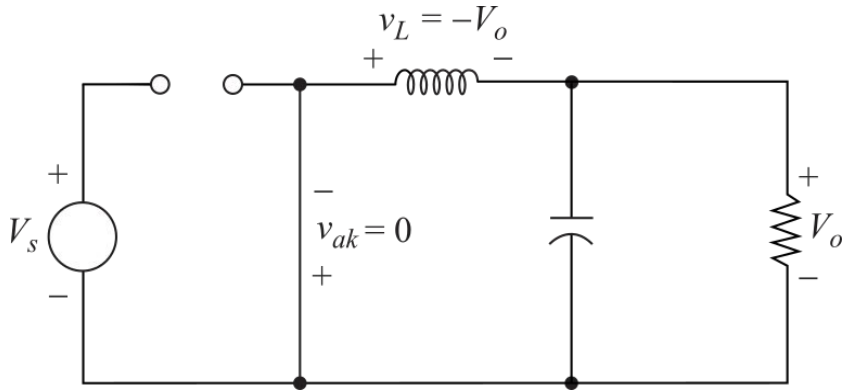
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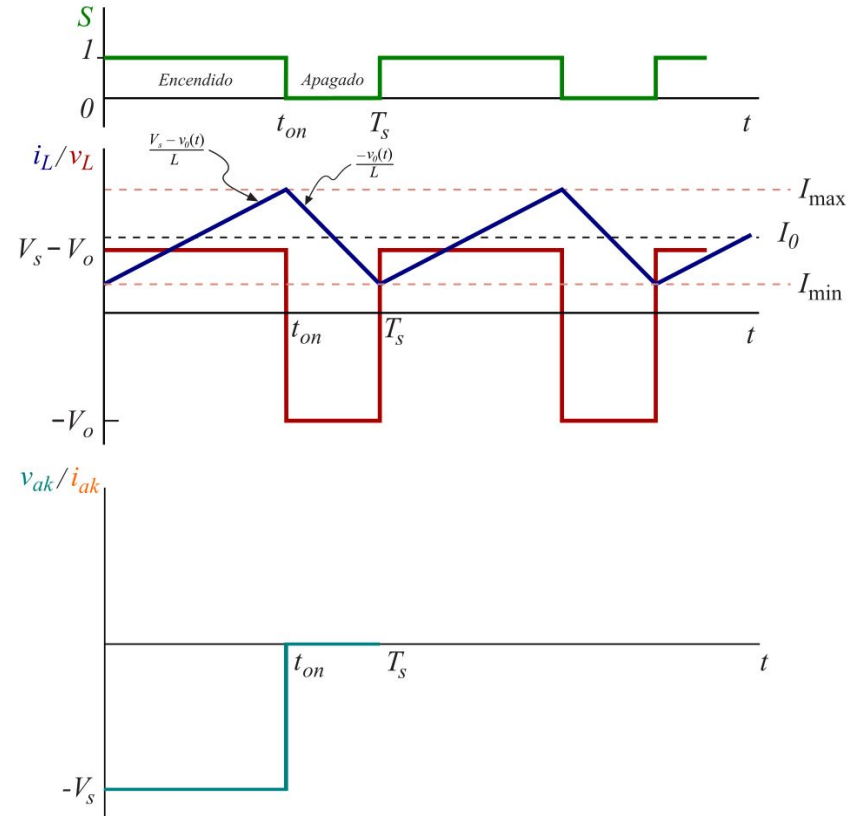
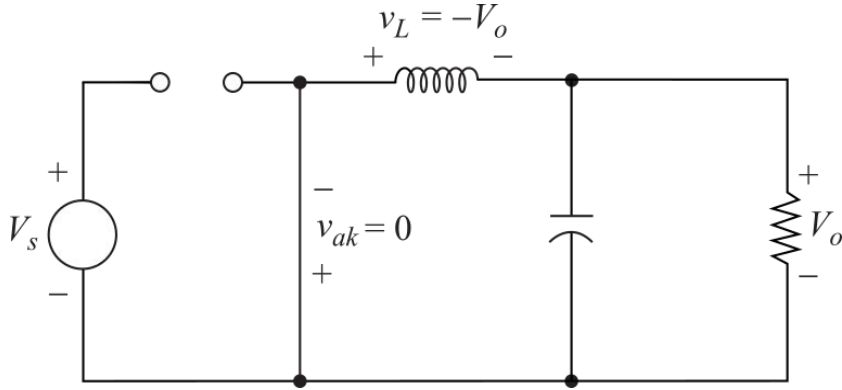
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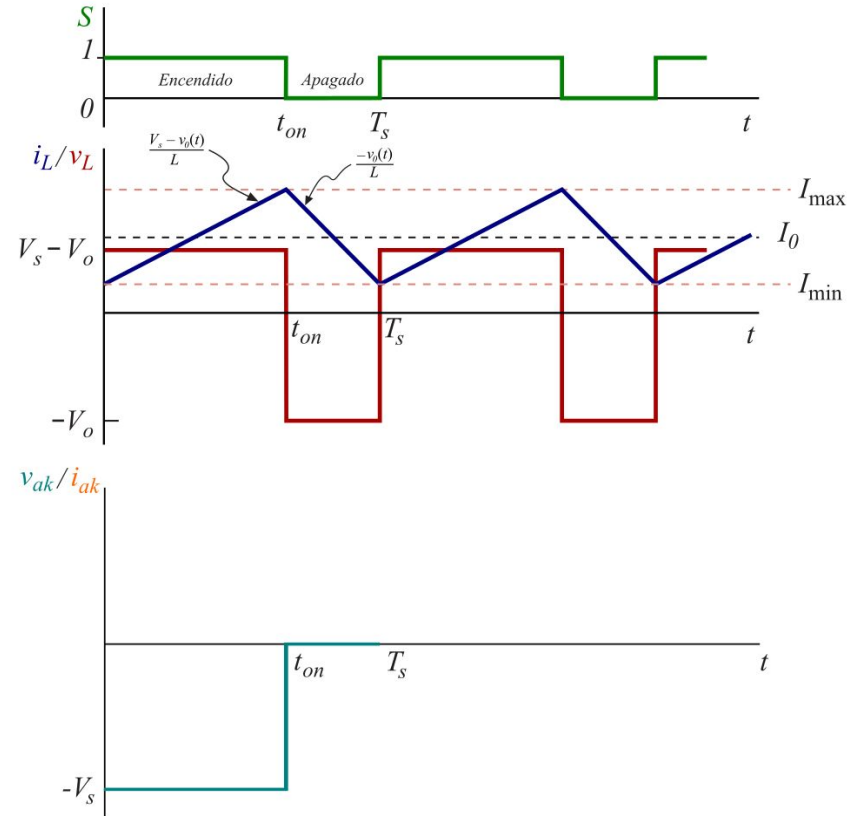
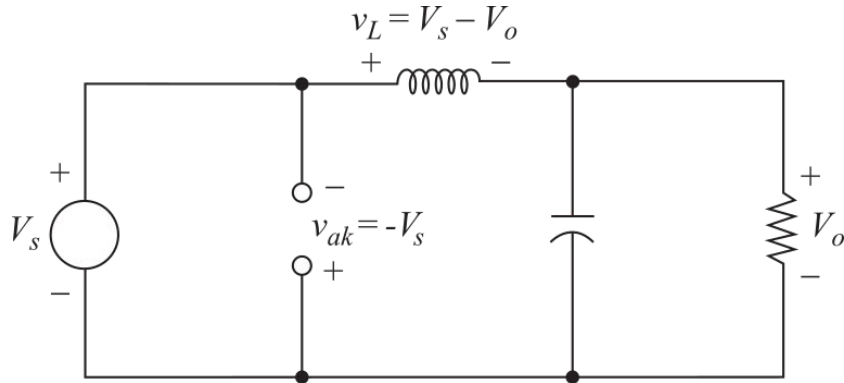
Convertidor Reductor: formas de onda



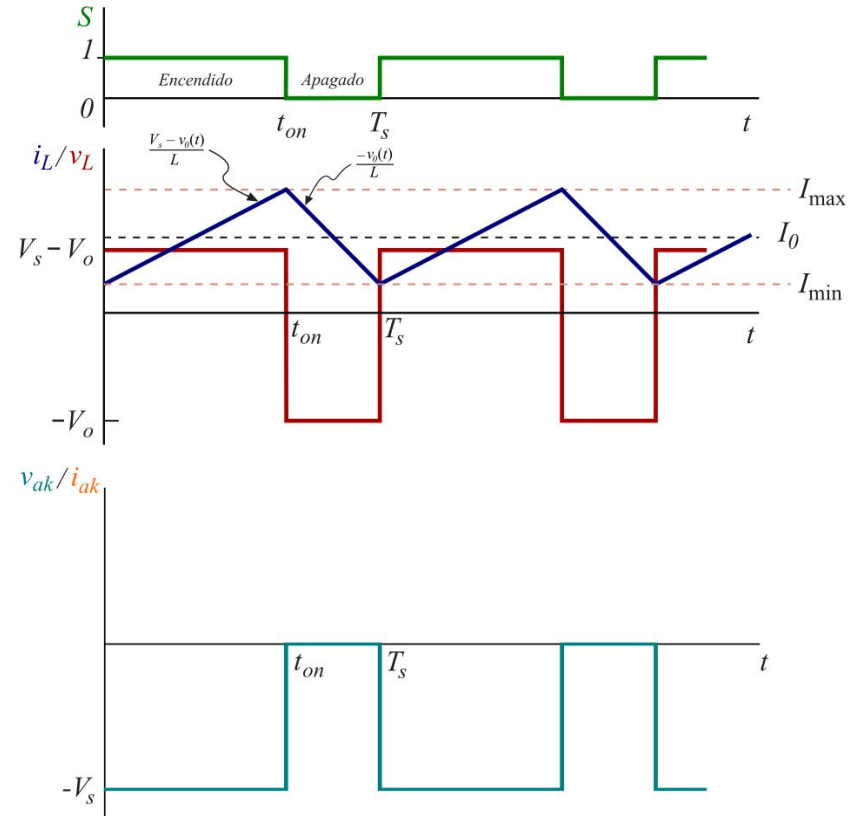
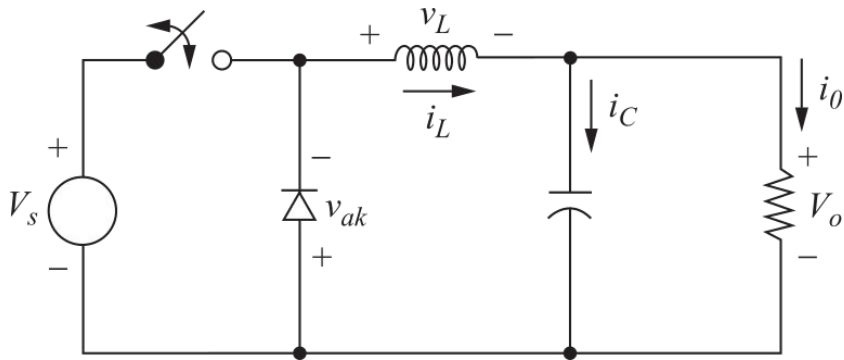
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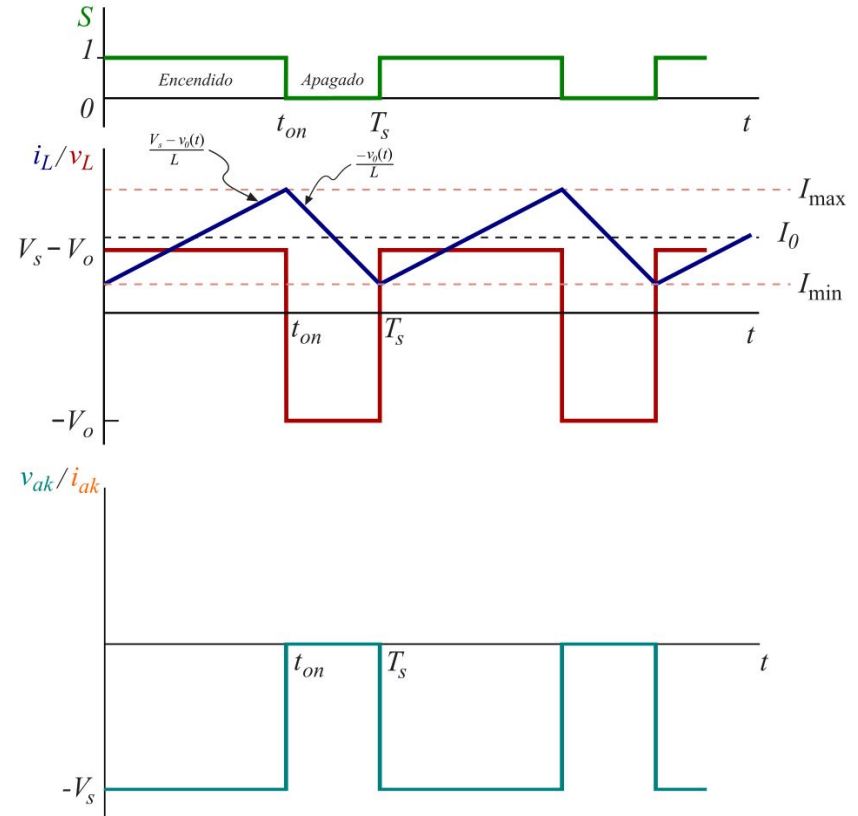
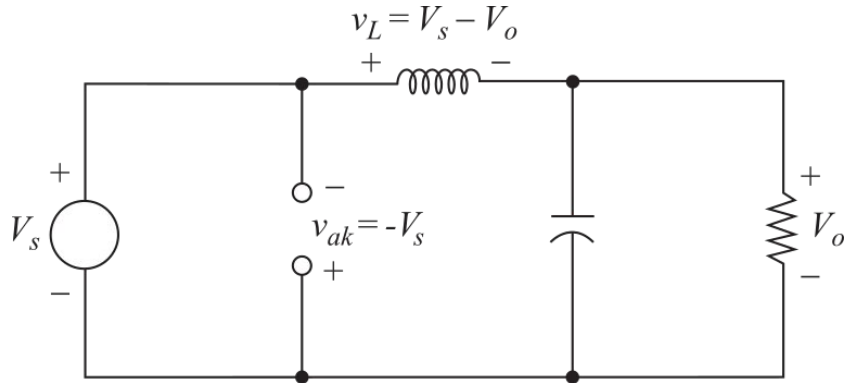
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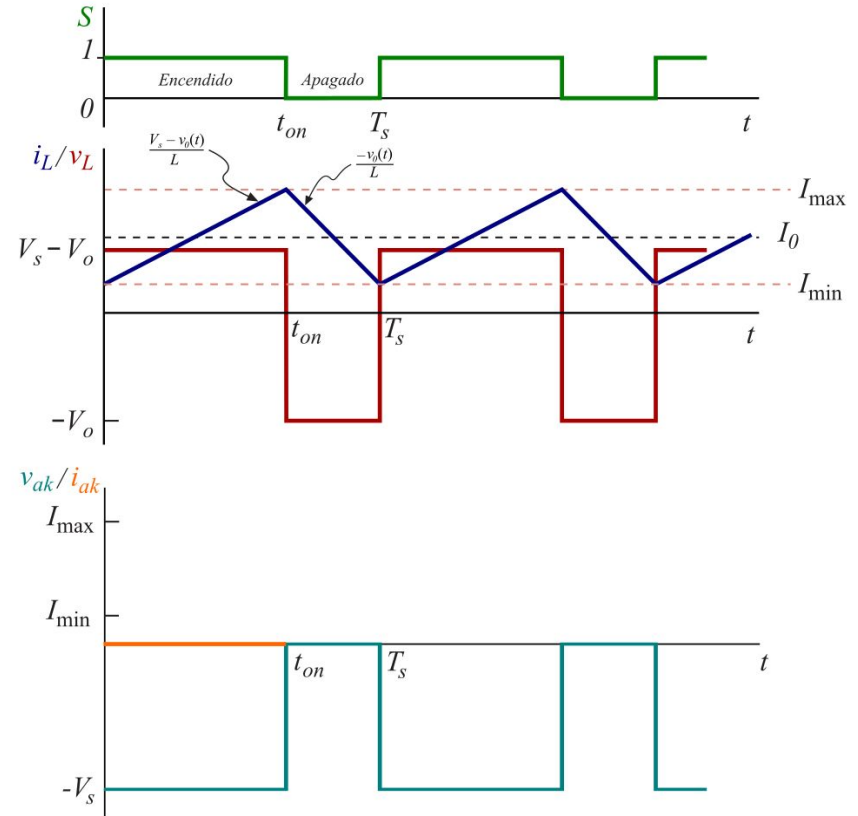
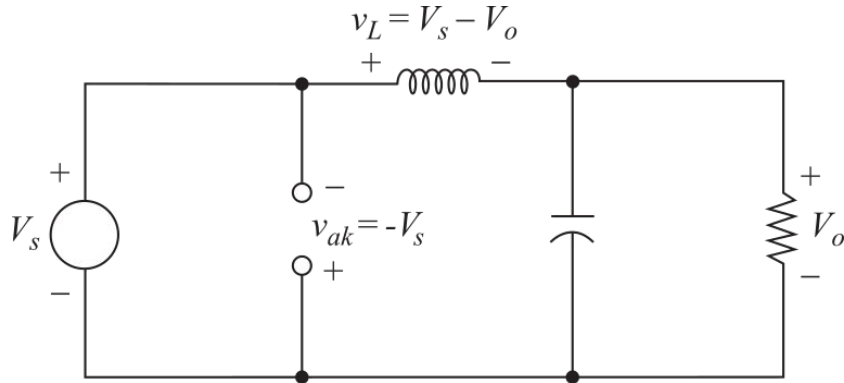
Convertidor Reductor: formas de onda



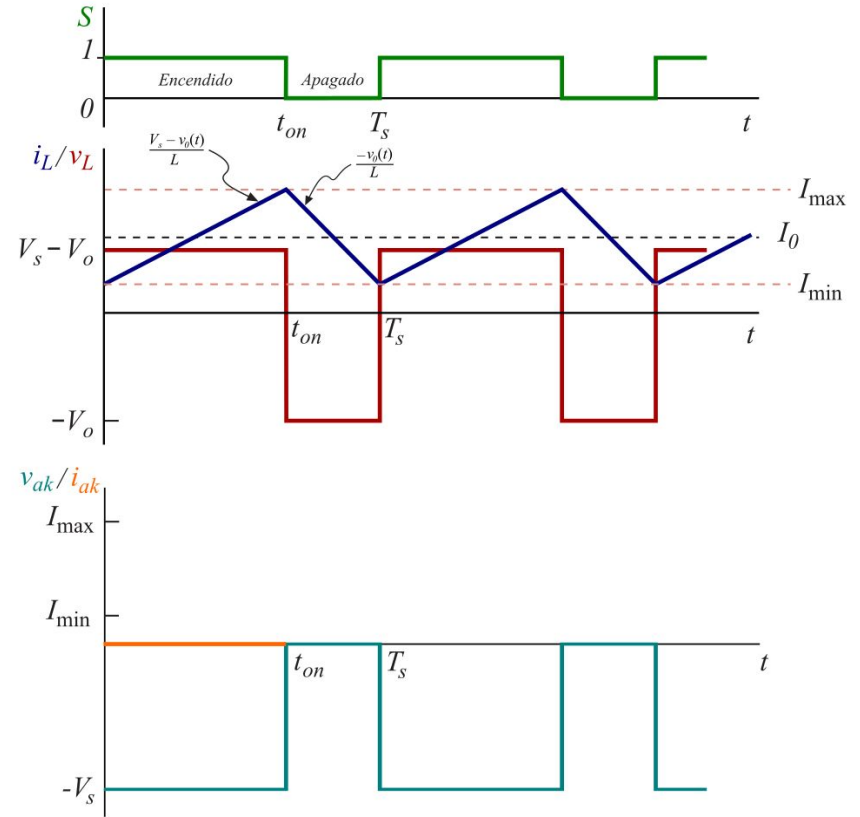
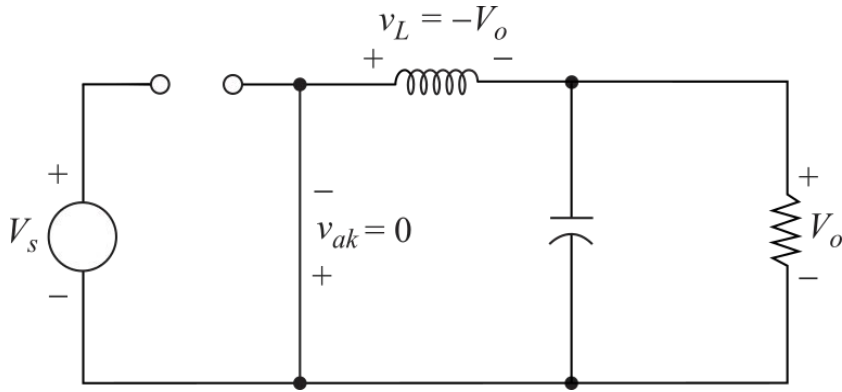
Convertidor Reductor: formas de onda



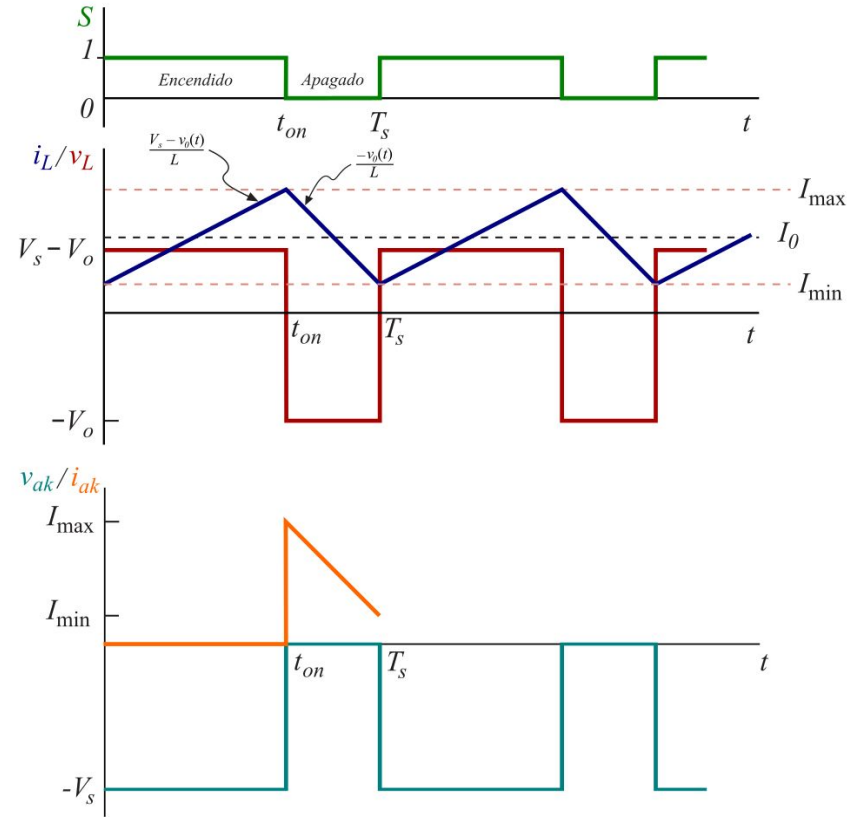
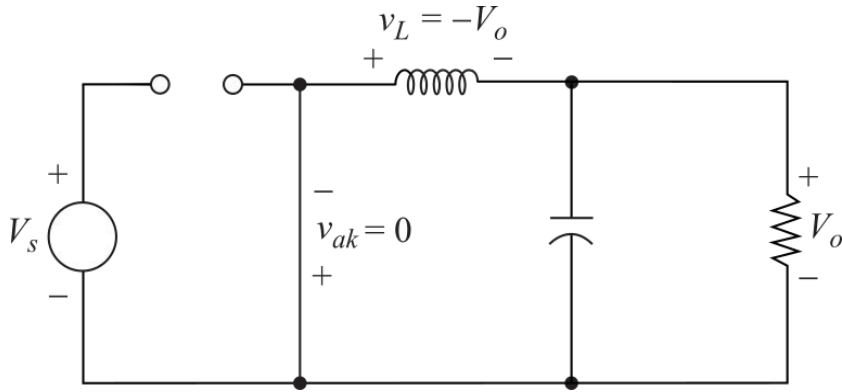
Convertidor Reductor: formas de onda



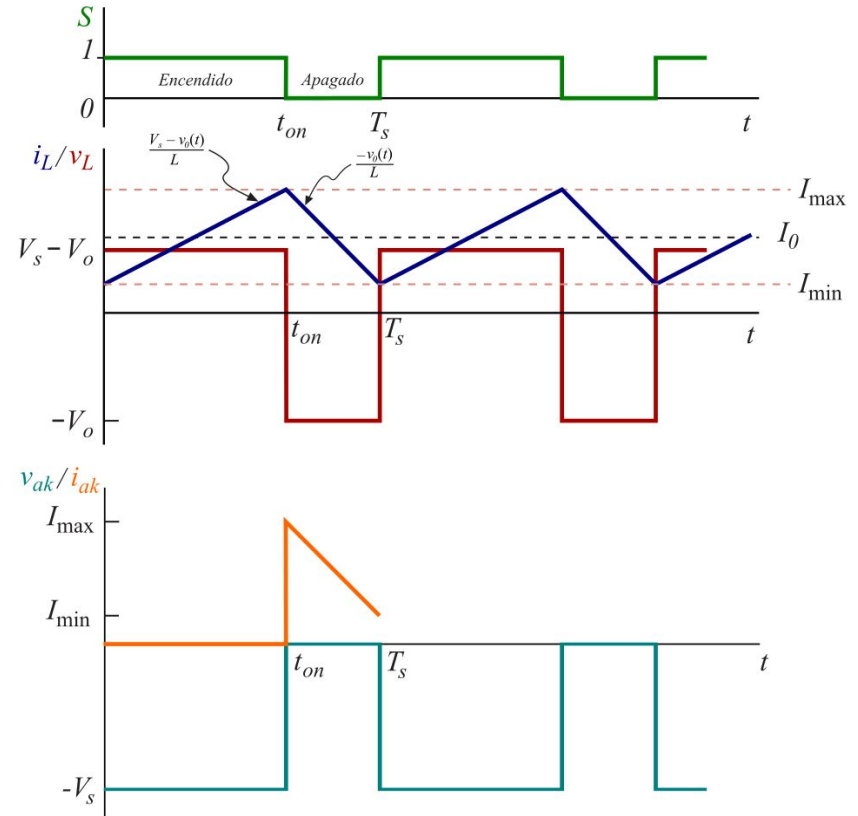
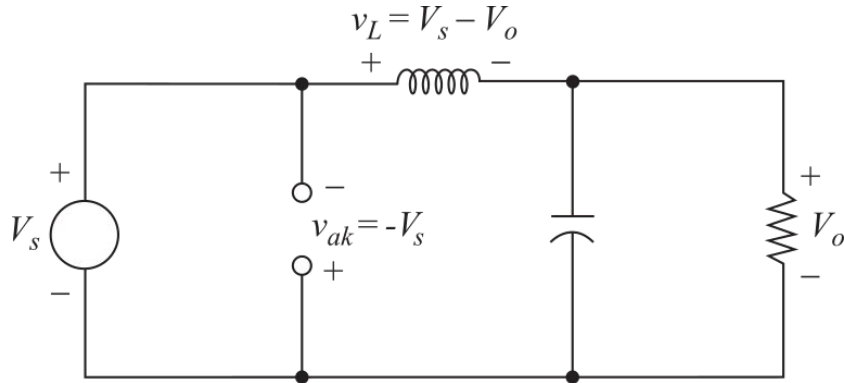
Convertidor Reductor: formas de onda



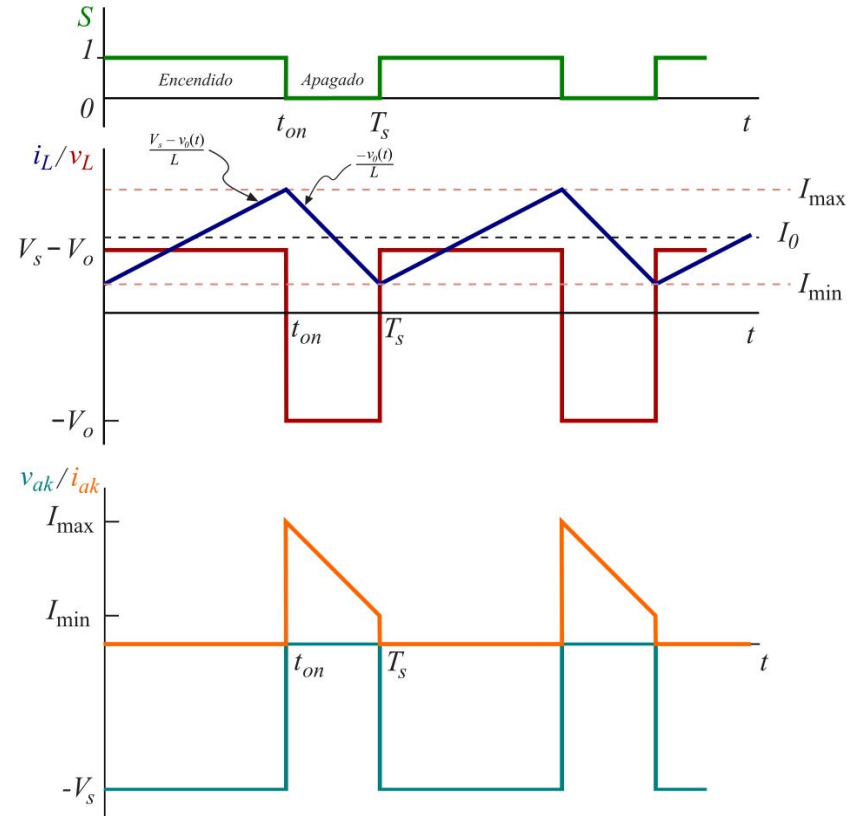
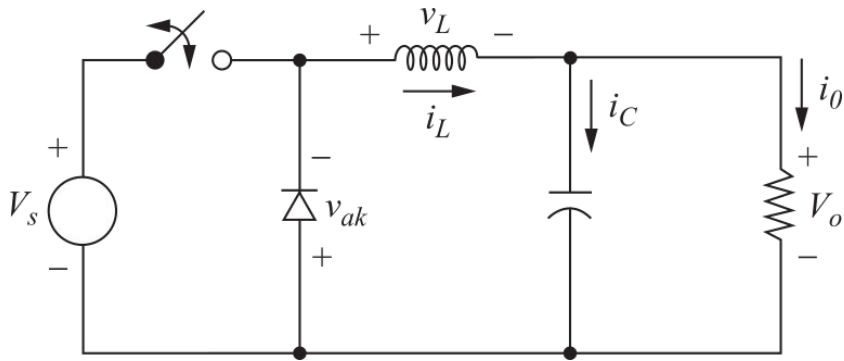
Convertidor Reductor: formas de onda



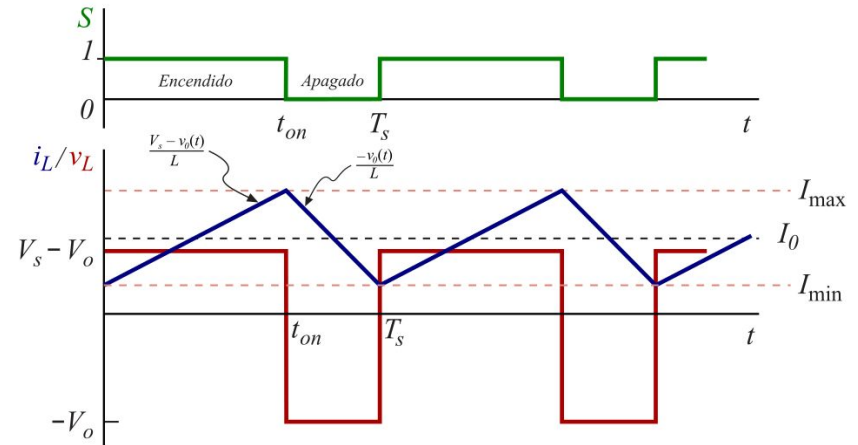
Convertidor Reductor: formas de onda



Convertidor Reductor: formas de onda

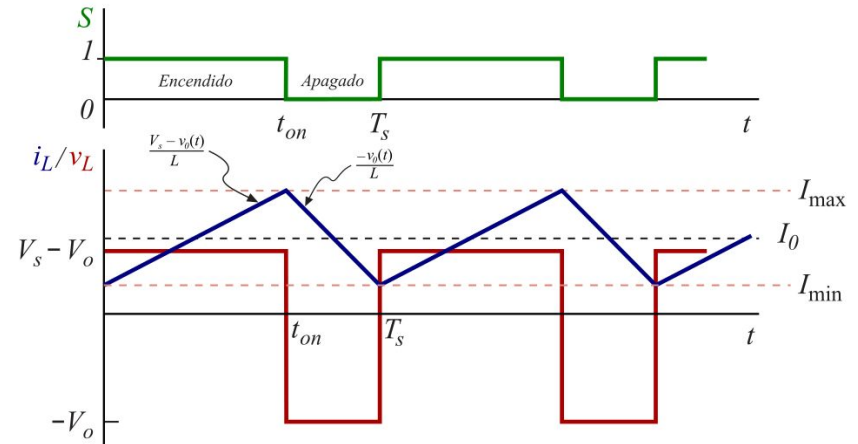


Convertidor Reductor: relación de conversión



Convertidor Reductor: relación de conversión

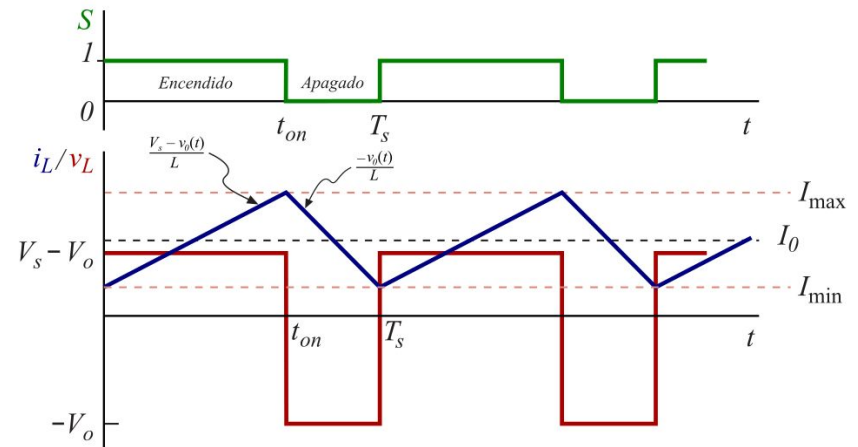
La tensión media en el inductor debe ser nula



Convertidor Reductor: relación de conversión

La tensión media en el inductor debe ser nula, entonces:

$$(V_s - V_o)t_{on} = V_o(T_s - t_{on})$$

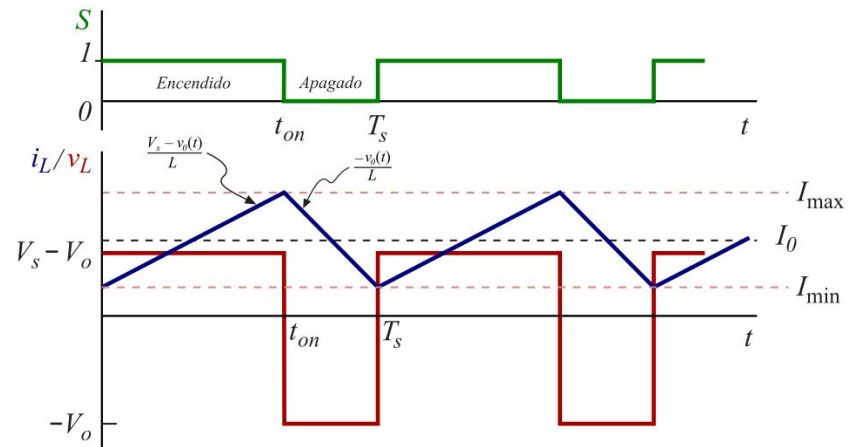


Convertidor Reductor: relación de conversión

La tensión media en el inductor debe ser nula, entonces:

$$(V_s - V_o)t_{on} = V_o(T_s - t_{on})$$

$$V_s t_{on} - V_o t_{on} = V_o T_s - V_o t_{on}$$

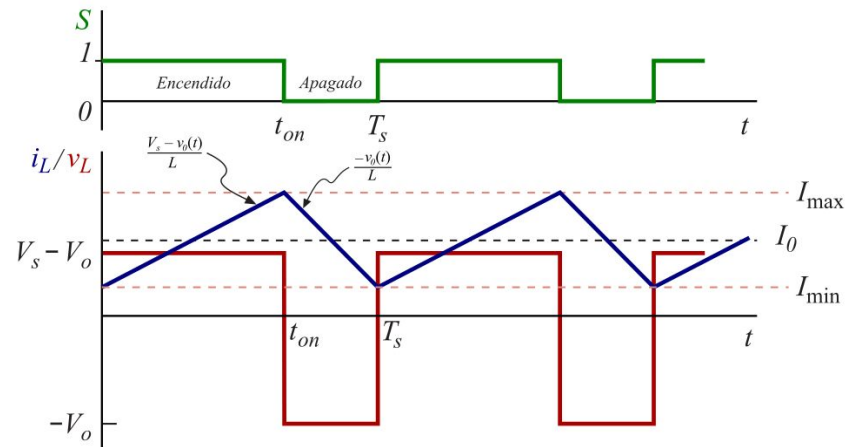


Convertidor Reductor: relación de conversión

La tensión media en el inductor debe ser nula, entonces:

$$(V_s - V_o)t_{on} = V_o(T_s - t_{on})$$

$$V_s t_{on} - \cancel{V_o t_{on}} = V_o T_s - \cancel{V_o t_{on}}$$



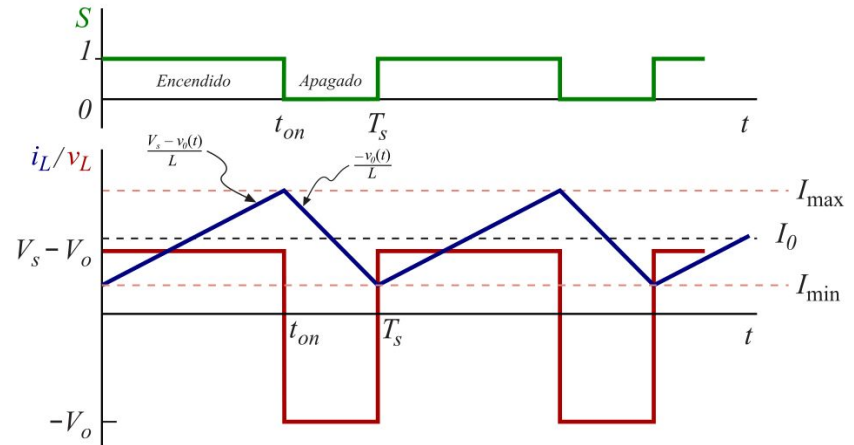
Convertidor Reductor: relación de conversión

La tensión media en el inductor debe ser nula, entonces:

$$(V_s - V_o)t_{on} = V_o(T_s - t_{on})$$

$$V_s t_{on} - \cancel{V_o t_{on}} = V_o T_s - \cancel{V_o t_{on}}$$

$$V_o = \frac{t_{on}}{T_s} V_s$$



Convertidor Reductor: relación de conversión

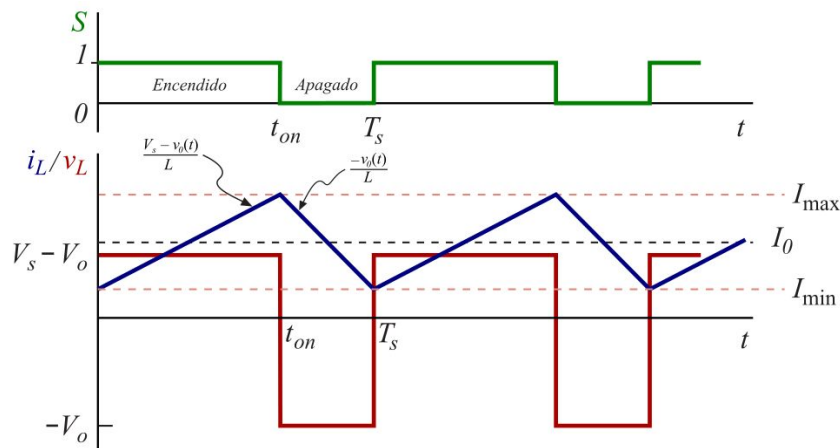
La tensión media en el inductor debe ser nula, entonces:

$$(V_s - V_o)t_{on} = V_o(T_s - t_{on})$$

$$V_s t_{on} - \cancel{V_o t_{on}} = V_o T_s - \cancel{V_o t_{on}}$$

$$V_o = \frac{t_{on}}{T_s} V_s$$

$$V_o = DV_s$$



Convertidor Reductor: relación de conversión

La tensión media en el inductor debe ser nula, entonces:

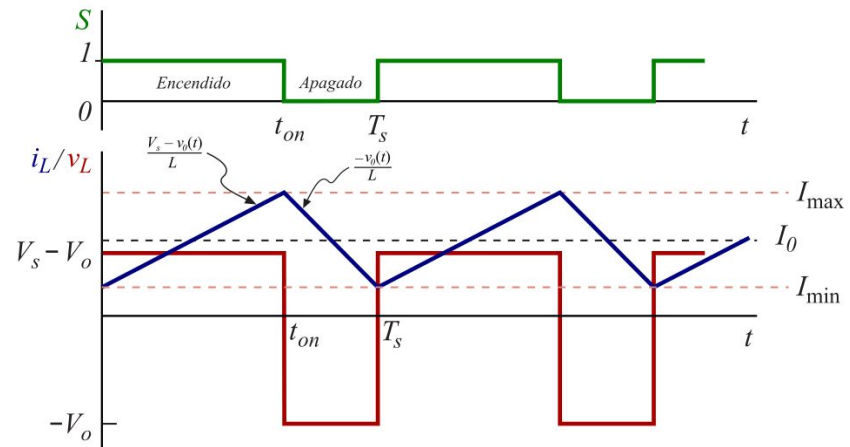
$$(V_s - V_o)t_{on} = V_o(T_s - t_{on})$$

$$V_s t_{on} - \cancel{V_o t_{on}} = V_o T_s - \cancel{V_o t_{on}}$$

$$V_o = \frac{t_{on}}{T_s} V_s$$

$$V_o = DV_s$$

$$0 \leq t_{on} \leq T_s$$



Convertidor Reductor: relación de conversión

La tensión media en el inductor debe ser nula, entonces:

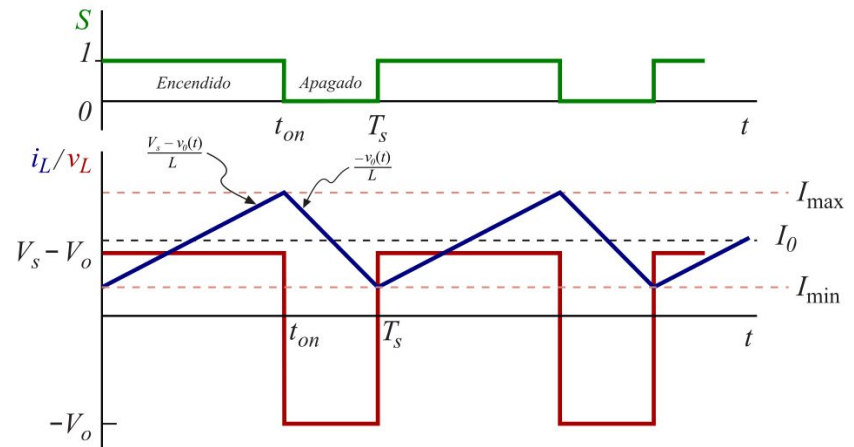
$$(V_s - V_o)t_{on} = V_o(T_s - t_{on})$$

$$V_s t_{on} - \cancel{V_o t_{on}} = V_o T_s - \cancel{V_o t_{on}}$$

$$V_o = \frac{t_{on}}{T_s} V_s$$

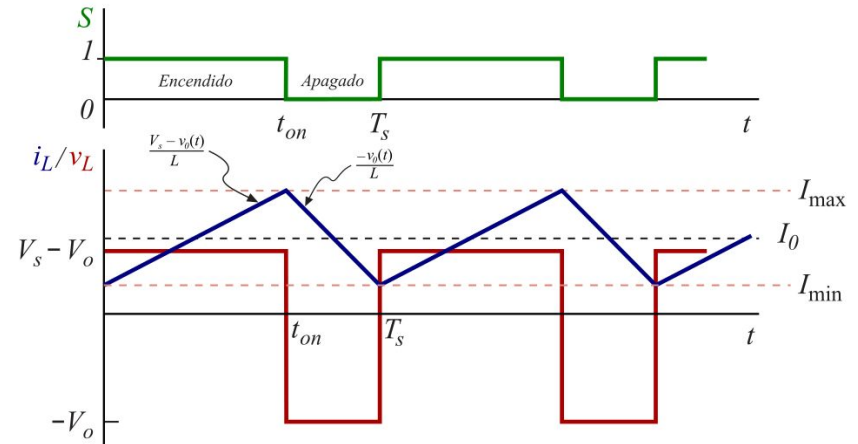
$$V_o = DV_s$$

$$0 \leq t_{on} \leq T_s \quad \longrightarrow \quad 0 \leq D \leq 1$$



Convertidor Reductor: relación de conversión

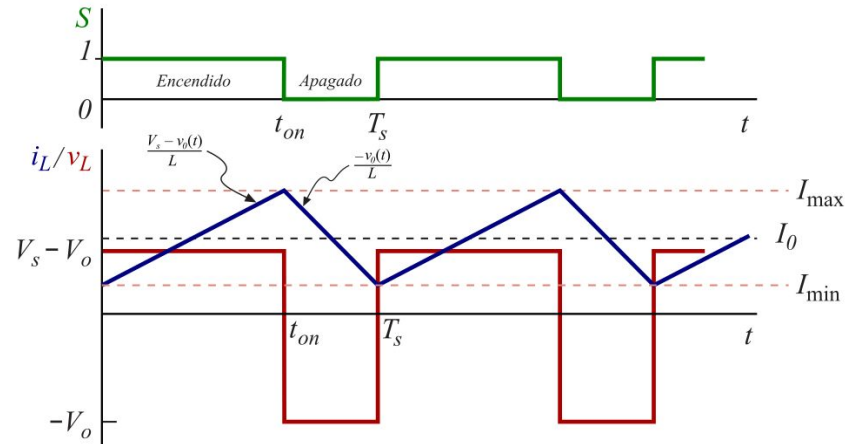
¿Qué pasa con la relación de corrientes?



Convertidor Reductor: relación de conversión

¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

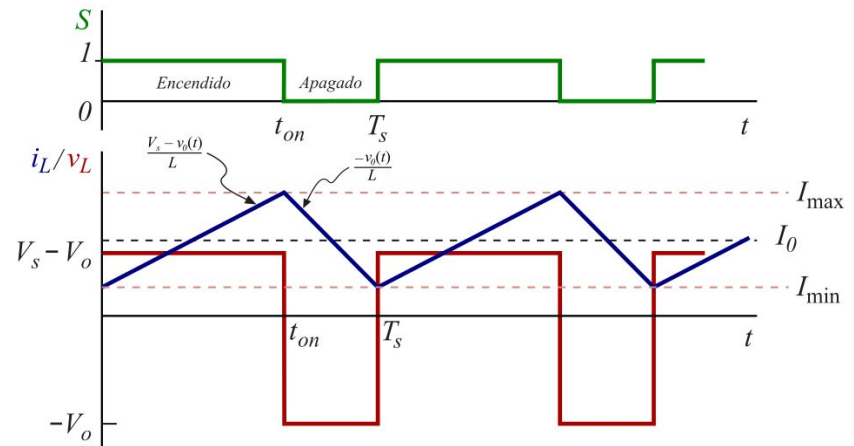


Convertidor Reductor: relación de conversión

¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

$$V_s I_s = V_o I_o$$

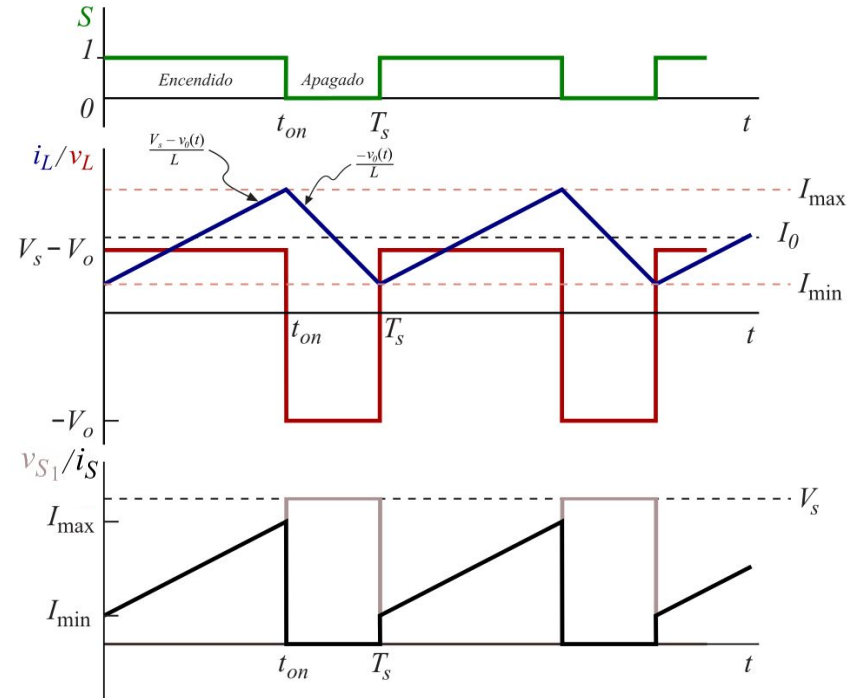


Convertidor Reductor: relación de conversión

¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

$$V_s I_s = V_o I_o$$



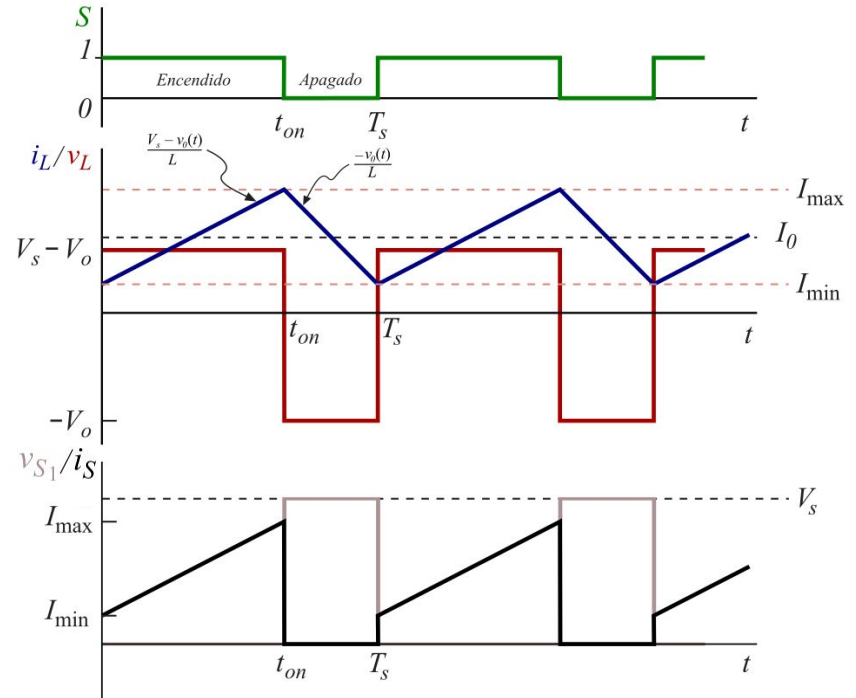
Convertidor Reductor: relación de conversión

¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

$$V_s I_s = V_o I_o$$

$$\frac{I_s}{I_o} = \frac{V_o}{V_s}$$



Convertidor Reductor: relación de conversión

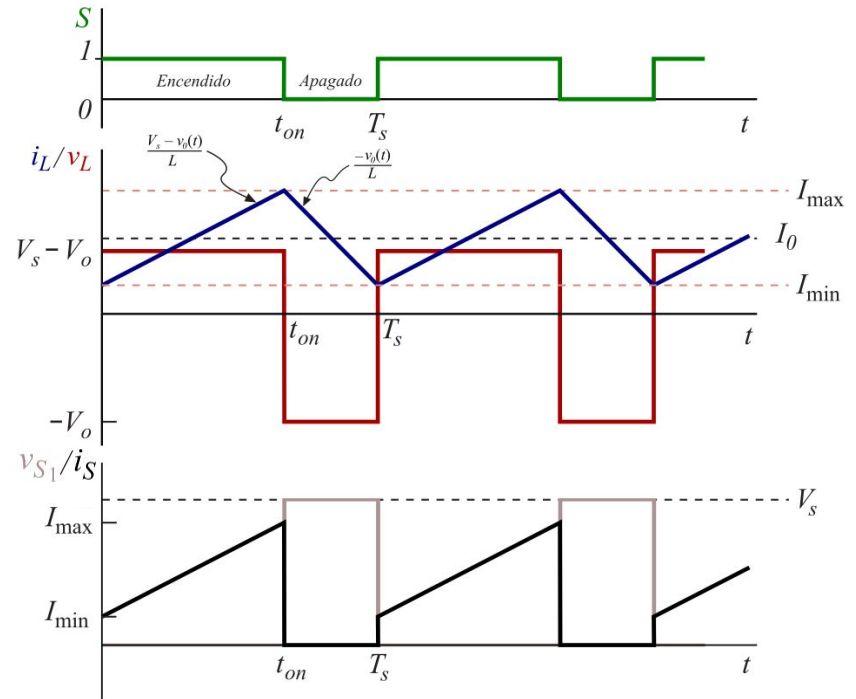
¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

$$V_s I_s = V_o I_o$$

$$\frac{I_s}{I_o} = \frac{V_o}{V_s}$$

$$V_o = D V_s$$



Convertidor Reductor: relación de conversión

¿Qué pasa con la relación de corrientes?

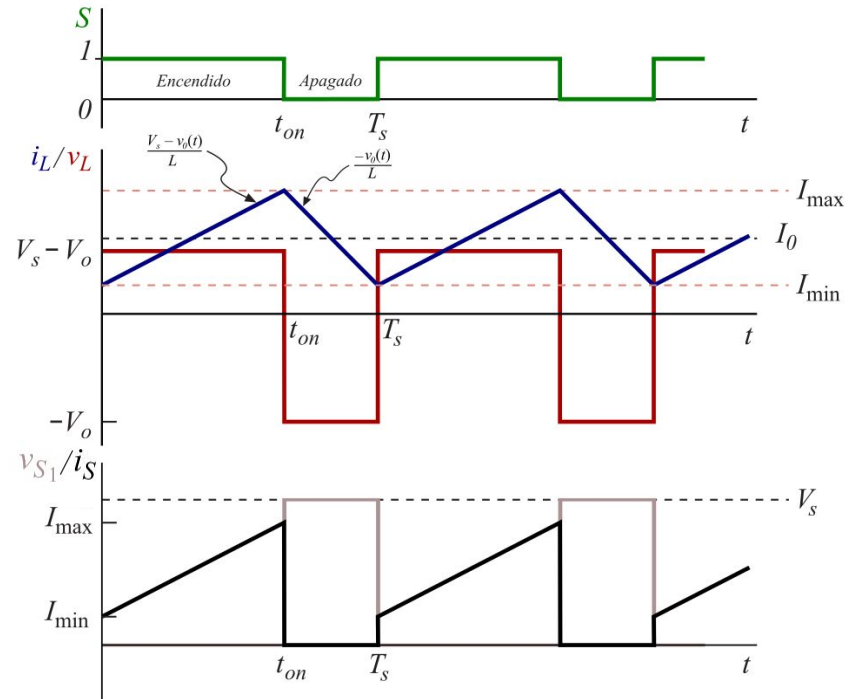
$$P_s = P_o$$

$$V_s I_s = V_o I_o$$

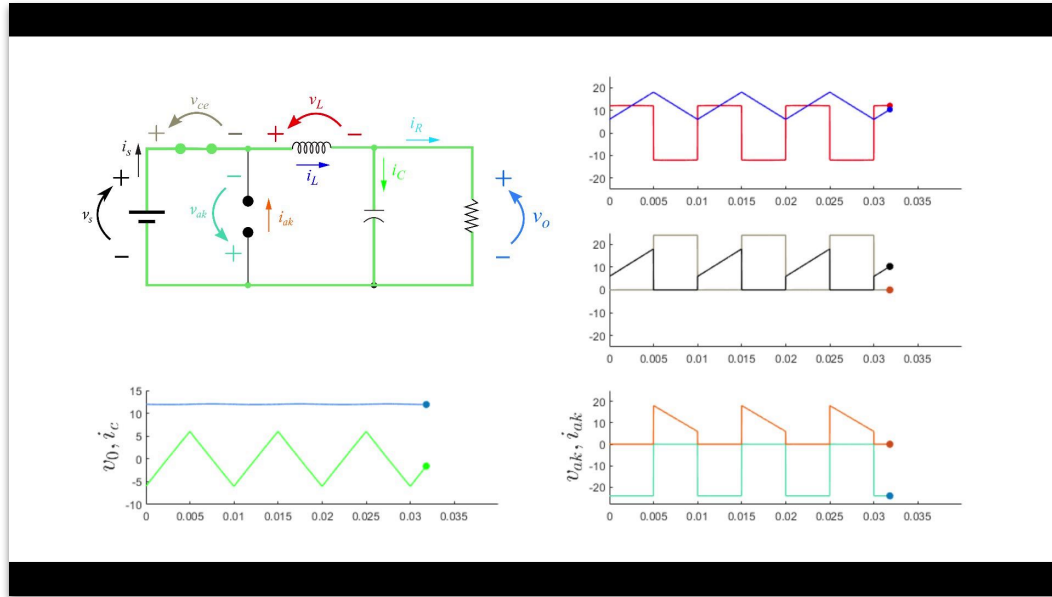
$$\frac{I_s}{I_o} = \frac{V_o}{V_s}$$

$$V_o = D V_s$$

$$I_s = D I_o$$



Convertidor Reductor: formas de onda



<https://youtu.be/SJ5WgmVTSPY>



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